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- 0-base / 1-base
- 線段樹改值懶標初始值不能設為 0
- DFS 的時候不小心覆寫到全域變數
- 多筆測資不能沒讀完直接 return
- vector 超級肥，小 vector 請用 array，例如矩陣快速冪

2 Init (Linux)

2.1 vimrc

```
1 syn on
2 set nu rnu cul mouse=a cin et ts=2 sw=2 sts=2 autochdir
  cb=unnamedplus
3 no ; :
4 ino {<CR> {<CR>}<Esc>ko
5 ca Hash w !cpp -fpreprocessed -dD -P \\\ tr -d '[:space:]' \\\
  openssl sha256 -r \\\ cut -c 6
1" Gino's .vimrc "
2"no <C-h> ^ | <C-l> $ | h b | l e | b l | e h
2.2 bashrc c08ff4
```

```
1 gogo() {
2   g++ -std=c++20 -O2 "$1" && echo run && ./a.out
3}
4 alias vig='vim -u ~/.vimrc-gino'
```

3 Basic

3.1 Template (Using Codebook)

```
1 // CryptoGrapheR
2 #define SZ(c) ((int)(c).size())
3 // kactl
4 #define sz(x) (int)(x).size()
5 #define rep(i, a, b) for(int i=(a); i<(b); i++)
6 using ll = long long;
7 using vi = vector<int>;
8 using pii = pair<int, int>;
9 // ckiseki
10 #define all(x) begin(x), end(x)
11 #define pb emplace_back
12 #define REP(i, l, r) for (int i=(l); i<=(r); ++i)
13 const ll LINF = 1e18;
14 template<class A, class B> bool chmin(A& a, B& b) {
15   return b < a ? a = b, 1 : 0;
16 }
```

3.2 PBDS, Random

```
1 #include <bits/extc++.h>
2 using namespace __gnu_pbds;
3 // map
4 tree<int, int, less<>, rb_tree_tag,
  tree_order_statistics_node_update> tr;
5 tr.order_of_key(element);
6 tr.find_by_order(rank);
7 // set
8 tree<int, null_type, less<>, rb_tree_tag,
  tree_order_statistics_node_update> tr;
9 tr.order_of_key(element);
10 tr.find_by_order(rank);
11 // priority queue
12 __gnu_pbds::priority_queue<int, less<int>> big_q;
13 __gnu_pbds::priority_queue<int, greater<int>> small_q;
14 ql.join(q2); // join
15
16 mt19937
  gen(chrono::steady_clock::now().time_since_epoch().count());
17 uniform_int_distribution<int>(a, b)(rng) // [a, b]
18 uniform_real_distribution<double>(a, b)(rng) // (a, b)
19 shuffle(v.begin(), v.end(), gen);
```

3.3 Debug

```
1 #define debug(...) cerr << "\x1b[33m" #_VA_ARGS__ " :
  \x1b[0m", \
2   [](auto&&... a){ ((cerr << a << ' '), ...); }
  (_VA_ARGS__), cerr << '\n'
3 #define output(...) \
4   [](auto&&... a){ ((cout << a << ' '), ...); }
  (_VA_ARGS__), cout << '\n'
5 int v = 42;
6 format("{:b}", v); // "101010"
7 format("{:#b}", v); // "0b101010"
8 format("{:x}", v); // "0x2a"
9 format("{:X}", v); // "0X2A"
10 int x = 7;
11 format("{:05d}", x); // "00007"
12 format("{:>5}", x); // " 7"
13 format("{:<5}", x); // "7 "
14 format("{:~5}", x); // " 7 "
15 format("{:*^5}", x); // "***7*"
16 double pi = 3.1415926535;
17 format("{:.2f}", pi); // "3.14"
```

1 Reminder

1.1 Observations and Tricks

- Contribution Technique
- 二分圖/Spanning Tree/DFS Tree
- 行、列操作互相獨立
- 奇偶性
- 當 s, t 遞增並且 $t = f(s)$ ，對 s 二分搜不好做，可以改成對 t 二分搜，再算 $f(t)$
- 啟發式合併
- Permutation Normalization (做一些平移對齊兩個 permutation)
- 枚舉 $a_1 \sim a_n$ 再枚舉 $a_n \sim a_1$ 可以包在一個迴圈
- 兩個凸型函數相加還是凸型函數，相減不一定
- 一個區間的 $\text{mex} = k$ ，表示這個區間包含 $0 \sim k-1$ 所有數字，並且「 U -區間」的最小值 $= k$ 。

1.2 Bug List

- 沒開 long long
- 陣列戳出界／陣列開不夠大
- 寫好的函式 (例如 `init()`、`build()`) 忘記呼叫

3.4 SVG Writer

```
1 // Author: ckiseki
2 class SVG {
3     void p(string_view s) { o << s; }
4     void p(string_view s, auto v, auto... vs) {
5         auto i = s.find('$');
6         o << s.substr(0, i) << v, p(s.substr(i + 1), vs...);
7     }
8     ofstream o; string c = "red";
9 public:
10    SVG(auto f, auto x1, auto y1, auto x2, auto y2) : o(f) {
11        p("<svg xmlns='http://www.w3.org/2000/svg' "
12          "viewBox='$ $ $ $'>\n"
13          "<style>{*stroke-width:0.5%;}</style>\n",
14          x1, -y2, x2 - x1, y2 - y1); }
15    ~SVG() { p("</svg>\n"); }
16    void color(string nc) { c = nc; }
17    void line(auto x1, auto y1, auto x2, auto y2) {
18        p("<line x1='$' y1='$' x2='$' y2='$' stroke='$'>\n",
19          x1, -y1, x2, -y2, c); }
20    void circle(auto x, auto y, auto r) {
21        p("<circle cx='$' cy='$' r='$' stroke='$' "
22          "fill='none'>\n", x, -y, r, c); }
23    void text(auto x, auto y, string s, int w = 12) {
24        p("<text x='$' y='$' font-size='px'>$</text>\n",
25          x, -y, w, s); }
26};

1 SVG svg("out.svg", -200, -200, 200, 200);
2 svg.color("lightgray"); // ↓ drawing x-y plane
3 svg.line(-200, 0, 200, 0); svg.line(0, -200, 0, 200);
4 svg.color("blue"); svg.line(10, 10, 80, 80);
5 svg.color("red"); svg.circle(50, 50, 1);
```

3.5 Python

b16508

```
1 import sys
2 input = sys.stdin.readline
3 readStr = lambda: input().rstrip('\r\n')
4 readInt = lambda: int(input())
5 readInts = lambda: list(map(int, input().split()))
6 from decimal import *
7 getcontext().prec = 50 # precision
8 x = Decimal(str(x))
9 y = Decimal("6.7")
10 x *= y
11 print(x.quantize(Decimal("0.000001"),
12   rounding=ROUND_HALF_EVEN))
13 # ROUND_CEILING : To INF (2.9 -> 3, -2.1 -> -2)
14 # ROUND_FLOOR : To -INF (2.1 -> 2, -2.9 -> -3)
15 # ROUND_HALF_EVEN: To nearest even (2.5 -> 2, 3.5 -> 4)
16 # ROUND_HALF_UP : Half away from 0 (2.5 -> 3, -2.5 -> -3)
17 # ROUND_HALF_DOWN: Half towards 0 (2.5 -> 2, -2.5 -> -2)
18 # ROUND_UP : Away from 0 (2.1 -> 3, -2.1 -> -3)
19 # ROUND_DOWN : Truncate to 0 (2.9 -> 2, -2.9 -> -2)
20 # <Default>: ROUND_HALF_EVEN
21 # <Standard School Math>: ROUND_HALF_UP
```

3.6 Stress Tests

```
1 from random import *
2 n = randint(1, 10)
3 arr = list(randint(1, 20) for i in range(n))
4 print(*arr)
5 # permutation / non-repetitive sequence
6 perm = sample(range(1, n + 1), n)
7 arr = sample(range(1, 10000 + 1), n)
8 # string
9 from string import *
10 s = "".join(choices(ascii_lowercase, k=n))
11 t = "".join(choices(ascii_lowercase[:3], k=n)) # (a|b|c)*
12 u = "".join(choices(ascii_lowercase + ascii_uppercase +
13   digits, k=n))
14 # palindrome
15 half = "".join(choices(ascii_lowercase, k=(n + 1) // 2))
16 palindrome = half + (half[-2::-1] if n % 2 else half[::-1])
17 # ===== tree =====
18 n = 30
19 ## shallow tree
20 edges = [(randint(1, u - 1), u) for u in range(2, n + 1)]
21 ## deep tree
22 (suggested k: 2 ~ 5 or int(sqrt(n)) or 2n/D (D is
23   expected depth))
24 from math import *
25 k = int(sqrt(n))
26 edges = [(randint(max(1, u - k), u - 1), u) for u in range(2, n
27   + 1)]
28 for e in edges:
29     print(*e)
30 edges = [(1, i) for i in range(2, n + 1)] ## star
31 edges = [(i, i + 1) for i in range(1, n)] ## chain
```

```
29 edges = [(i // 2, i) for i in range(2, n + 1)] ## complete
30 binary tree
31 # ===== graph =====
32 ## not necessarily connected, no multi-edge or self-loop
33 n = 10
34 density = 0.3 # graph density (0.0 ~ 1.0)
35 edges = [(i, j) for i in range(1, n + 1) for j in range(i + 1,
36   n + 1) if random() < density]
37 ## connected, no multi-edge or self-loop
38 n, extra_edges = 10, 5
39 edges = set()
40 ##### ... generate spanning tree with codes above
41 while len(edges) < n - 1 + extra_edges:
42     u, v = randint(1, n), randint(1, n)
43     if u != v:
44         edges.add((min(u, v), max(u, v)))
45 edges = list(edges)
46 ## DAG
47 perm = sample(range(1, n + 1), n)
48 edges = []
49 for i in range(m):
50     i, j = sorted(sample(range(n), 2))
51     edges.append((perm[i], perm[j]))
52 ## Bipartite Graph
53 n1, n2 = 6, 7
54 left, right = list(range(1, n1 + 1)), list(range(n1 + 1,
55   n1 + n2 + 1))
56 edges = [(choice(left), choice(right)) for _ in range(m)]
```

```
1 g++ -std=c++20 -o buggy $1
2 g++ -std=c++20 -o ac ac.cpp
3 for i in {1..100}; do
4     echo "TEST $i"
5     python3 gen.py > in
6     ./ac < in > oa
7     ./buggy < in > ob
8     if ! diff -q oa ob; then
9         cat in <(echo "=== AC") oa <(echo "=== WA") ob
10        break
11    fi
12 done
```

4 Data Structure

4.1 Mo's Algorithm

5308dd

```
1 // segments are 0-based
2 int B = max(1, (int)sqrt(n));
3 sort(Q.begin(), Q.end(), [&](const auto& a, const auto& b) {
4     if (a[0]/B != b[0]/B) return a[0] < b[0];
5     return (a[0]/B & 1) ? a[1] < b[1] : a[1] > b[1];
6 });
7 ll cur = 0; // current answer
8 int pl = 0, pr = -1;
9 for (auto& qi : Q) {
10    // get (l, r, qid) from qi
11    while (pl < l) del(pl++); while (pl > l) add(--pl);
12    while (pr < r) add(++pr); while (pr > r) del(pr--);
13    ans[qid] = cur;
14 }
```

4.2 CDQ

e1864b

```
1 // sorted by x; type 0 = update (weight w), 1 = query
2 // query: ans[id] = sum of w over updates with x' <= x,
3 // y' <= y, z' <= z (< if strict[dim]). 值域 [1, mxc]
4 // Dependency: 值域 BIT
5 struct Info { int x, y, z, type; ll w; int id; };
6 struct CDQ {
7     static bool cy(const Info& p, const Info& q)
8     { return p.y < q.y; }
9     vector<Info> a; int n; array<bool, 3> strict;
10    vector<ll> ans; BIT bit;
11    void dfs(int l, int r) { if (r - l <= 1) return;
12      int m = (l + r) / 2, i = l;
13      dfs(l, m);
14      vector<Info> t(a.begin() + m, a.begin() + r);
15      ranges::sort(t, cy);
16      for (auto& q : t) {
17          for (; i < m && a[i].y < q.y + !strict[1]; i++)
18              if (!a[i].type) bit.upd(a[i].z, a[i].w);
19          if (q.type)
20              ans[q.id] += bit.qry(q.z - strict[2]);
21      }
22      for (int k = l; k < i; k++)
23          if (!a[k].type) bit.upd(a[k].z, -a[k].w);
24      dfs(m, r);
25      inplace_merge(a.begin() + l, a.begin() + m,
26        a.begin() + r, cy);
27  }
28  CDQ(vector<Info> _a, int mxc, array<bool, 3> _s = {}):
29      a(_a), n(a.size()), strict(_s) {
30      ans.assign(n, 0); bit.init(mxc);
```

```

31 auto key = [&](Info& p) { return
32     make_tuple(p.x, p.type != strict[0], p.y, p.z); };
33 ranges::sort(a, {}, key); dfs(0, n);
34 }
35 };

```

4.3 Persistent Treap

5f64da

```

1 // Author: Ian
2 struct node {
3     node *l, *r;
4     char c; int v, sz;
5     node(char x = '$'): c(x), v(mt()), sz(1) {
6         l = r = nullptr;
7     }
8     node(node* p) { *this = *p; }
9     void pull() {
10         sz = 1;
11         for (auto i : {l, r})
12             if (i) sz += i->sz;
13     }
14 } arr[maxn], *ptr = arr;
15 inline int size(node* p) { return p ? p->sz : 0; }
16 node* merge(node* a, node* b) {
17     if (!a || !b) return a ? a : b;
18     if (a->v < b->v) {
19         node* ret = new(ptr++) node(a);
20         ret->r = merge(ret->r, b); ret->pull();
21         return ret;
22     }
23     else {
24         node* ret = new(ptr++) node(b);
25         ret->l = merge(a, ret->l); ret->pull();
26         return ret;
27     }
28 }
29 P<node*> split(node* p, int k) {
30     if (!p) return {nullptr, nullptr};
31     if (k >= size(p->l) + 1) {
32         auto [a, b] = split(p->r, k - size(p->l) - 1);
33         node* ret = new(ptr++) node(p);
34         ret->r = a, ret->pull();
35         return {ret, b};
36     }
37     else {
38         auto [a, b] = split(p->l, k);
39         node* ret = new(ptr++) node(p);
40         ret->l = b, ret->pull();
41         return {a, ret};
42     }
43 }

```

4.4 Li Chao Tree

110f70

```

1 // Author: Ian
2 // insert(L): add line L(ax+b) in [0, maxn)
3 // insertseg(L, ql, qr): add segment L in [ql, qr)
4 // query(x): find maximum L at x
5 using T=long long;
6 const int maxn = 1e5+5;
7 const T MN = -1e15;
8 struct line {
9     T a, b;
10    line(T a=0, T b=MN): a(a), b(b) {}
11    T operator()(T x) { return a * x + b; }
12 } arr[(maxn+1)<<3];
13 bool operator<(line a, line b) {
14     return a.a < b.a || (a.a == b.a && a.b < b.b); }
15 #define m ((l+r)>>1)
16 void insert(line x, int i=1, int l=0, int r=maxn) {
17     if (r-l == 1) {
18         if (x(l) > arr[i](l)) arr[i] = x;
19         return; }
20     line a = max(arr[i], x), b = min(arr[i], x);
21     if (a(m) > b(m))
22         arr[i] = a, insert(b, i<<1, l, m);
23     else
24         arr[i] = b, insert(a, i<<1|1, m, r);
25 }
26 void insertseg(line x, int ql, int qr,
27     int i=1, int l=0, int r=maxn) {
28     if (qr<=l || r<=ql || ql>=qr) return;
29     if (ql<=l && r<=qr) { insert(x, i, l, r); return; }
30     insertseg(x, ql, qr, i<<1, l, m);
31     insertseg(x, ql, qr, i<<1|1, m, r);
32 }
33 T query(T x, int i=1, int l=0, int r=maxn) {
34     if (x<l || r<=x) return MN;
35     if (r-l == 1) return arr[i](x);
36     return max({arr[i](x),
37         query(x, i<<1, l, m), query(x, i<<1|1, m, r)});

```

38 }

39 #undef m

4.5 Time Segment Tree

e8ac31

```

1 // Author: Ian
2 struct Op; // reversible operation
3 const int maxn = 2e5 + 5;
4 vector<Op> arr[(maxn + 1) << 2];
5 #define m ((l + r) >> 1)
6 void insert(int ql, int qr, Op x, int i=1, int l=0, int r=maxn) {
7     if (qr <= l || r <= ql) return;
8     if (ql <= l && r <= qr) { arr[i].push_back(x); return; }
9     if (qr <= m) insert(ql, qr, x, i << 1, l, m);
10    else if (m <= ql) insert(ql, qr, x, i << 1 | 1, m, r);
11    else {
12        insert(ql, qr, x, i << 1, l, m);
13        insert(ql, qr, x, i << 1 | 1, m, r);
14    }
15 void traversal(vector<int>& ans, int i=1, int l=0, int r=maxn) {
16     int opcnt = 0;
17     for (auto val : arr[i])
18         if (DO_SOMETHING(val)) opcnt++;
19     if (r - l == 1) ans[l] = CALC_ANS();
20     else {
21         traversal(ans, i << 1, l, m);
22         traversal(ans, i << 1 | 1, m, r);
23     }
24     while (opcnt--) UNDO_SOMETHING();
25     arr[i].clear();
26 }
27 #undef m

```

4.6 Segment Tree Beats

d7946d

```

1 // Author: Ian, https://judge.yosupo.jp/submission/400861
2 const int maxn=2e5+5;
3 const ll MX = 1e18+5;
4 const ll MN = -MX;
5 #define m ((l+r)>>1)
6 struct node {
7     ll sum, inc;
8     ll mn, smn; int mnc;
9     ll mx, smx; int mxc;
10    node(ll x=0): sum(x), inc(0),
11        mn(x), smn(MX), mnc(1),
12        mx(x), smx(MN), mxc(1) {}
13    void pull(int i);
14    void push(int i, int l, int r);
15 } a[(maxn+1)<<2];
16 void node::pull(int i) {
17     sum = a[i<<1].sum+a[i<<1|1].sum;
18     if (a[i<<1].mn == a[i<<1|1].mn) {
19         mn = a[i<<1].mn;
20         smn = min(a[i<<1].smn, a[i<<1|1].smn);
21         mnc = a[i<<1].mnc+a[i<<1|1].mnc;
22     } else {
23         if (a[i<<1].mn < a[i<<1|1].mn) {
24             mn = a[i<<1].mn;
25             smn = min(a[i<<1].smn, a[i<<1|1].smn);
26             mnc = a[i<<1].mnc;
27         } else {
28             mn = a[i<<1|1].mn;
29             smn = min(a[i<<1|1].smn, a[i<<1].smn);
30             mnc = a[i<<1|1].mnc;
31         }
32     }
33     if (a[i<<1].mx == a[i<<1|1].mx) {
34         mx = a[i<<1].mx;
35         smx = max(a[i<<1].smx, a[i<<1|1].smx);
36         mxc = a[i<<1].mxc+a[i<<1|1].mxc;
37     } else {
38         if (a[i<<1].mx > a[i<<1|1].mx) {
39             mx = a[i<<1].mx;
40             smx = max(a[i<<1].smx, a[i<<1|1].smx);
41             mxc = a[i<<1].mxc;
42         } else {
43             mx = a[i<<1|1].mx;
44             smx = max(a[i<<1|1].smx, a[i<<1].smx);
45             mxc = a[i<<1|1].mxc;
46         }
47     }
48 void updmn(int i, int len, ll v) {
49     a[i].sum += (v-a[i].mn)*a[i].mnc; a[i].mn = v;
50     if (len==1 || v>=a[i].mx)
51         a[i].mx = v, a[i].smx = MN;
52     else if (v>a[i].smx)
53         a[i].smx = v;
54 void updmx(int i, int len, ll v) {
55     a[i].sum -= (a[i].mx-v)*a[i].mxc; a[i].mx = v;
56     if (len==1 || v<=a[i].mn)
57         a[i].mn = v, a[i].smn = MX;

```

```

58 else if (v<a[i].smn)
59     a[i].smn = v;
60 }
61 void node::push(int i, int l, int r) {
62     if (r-l==1) return;
63     pair<int, int> ch[] = {{i<<1, m-l}, {i<<1|1, r-m}};
64     if (inc!=0) for (auto [j, len]: ch) {
65         a[j].sum += inc*len, a[j].inc += inc;
66         a[j].mx += inc, a[j].smx += inc;
67         a[j].mn += inc, a[j].smn += inc;
68     } inc = 0;
69     for (auto [j, len]: ch) {
70         if (a[j].mn < mn) updmn(j, len, mn);
71         if (a[j].mx > mx) updmx(j, len, mx); }
72 }
73 void build(vector<ll>& v, int i=1, int l=0, int r=maxn) {
74     if ((int)v.size()<=l) return;
75     if (r-l==1) { a[i] = node(v[l]); return; }
76     build(v, i<<1, l, m);
77     build(v, i<<1|1, m, r);
78     a[i].pull(i);
79 }
80 void chmin(int ql, int qr, ll v, int i=1, int l=0, int r=maxn) {
81     if (qr<=l || r<=ql || a[i].mx<=v) return;
82     if (ql<=l && r<=qr && a[i].smx<v) { updmx(i, r-l, v); return; }
83     a[i].push(i, l, r);
84     chmin(ql, qr, v, i<<1, l, m);
85     chmin(ql, qr, v, i<<1|1, m, r);
86     a[i].pull(i);
87 }
88 void chmax(int ql, int qr, ll v, int i=1, int l=0, int r=maxn) {
89     if (qr<=l || r<=ql || a[i].mn>=v) return;
90     if (ql<=l && r<=qr && a[i].smn>v) { updmn(i, r-l, v); return; }
91     a[i].push(i, l, r);
92     chmax(ql, qr, v, i<<1, l, m);
93     chmax(ql, qr, v, i<<1|1, m, r);
94     a[i].pull(i);
95 }
96 void inc(int ql, int qr, ll v, int i=1, int l=0, int r=maxn) {
97     if (qr<=l || r<=ql) return;
98     if (ql<=l && r<=qr) {
99         a[i].sum += v*(r-l);
100        a[i].mx += v, a[i].smx += v;
101        a[i].mn += v, a[i].smn += v;
102        a[i].inc += v;
103        return;
104    }
105    a[i].push(i, l, r);
106    inc(ql, qr, v, i<<1, l, m);
107    inc(ql, qr, v, i<<1|1, m, r);
108    a[i].pull(i);
109 }
110 ll qsum(int ql, int qr, int i=1, int l=0, int r=maxn) {
111     if (qr<=l || r<=ql) return 0;
112     if (ql<=l && r<=qr) return a[i].sum;
113     a[i].push(i, l, r);
114     return qsum(ql, qr, i<<1, l, m) + qsum(ql, qr, i<<1|1, m, r);
115 }
116 #undef m

```

4.7 Range Affine SegT: Metadata

32a905

```

1 // Author: Gino, Range Affine Segment Tree
2 // Maintain b*a[i]+c, support range modifications of b, c
3 // ===== RANGE MAX/MIN =====
4 struct S { ll mx; };
5 struct T { ll b, c; }; // bx + c, b >= 0
6 inline S op(S a, S b) { return {max(a.mx, b.mx)}; }
7 inline S e() { return {LLONG_MIN}; } // min: LLONG_MAX
8 inline S mapping(T f, S x) {
9     if (x.mx == LLONG_MIN) return x; // keep -inf
10    return {f.b * x.mx + f.c}; }
11 inline T composition(T f, T g) {
12     return {f.b * g.b, f.b * g.c + f.c}; }
13 }
14 inline T id() { return {1, 0}; }
15 // ===== RANGE SUM =====
16 struct S { ll x, len; };
17 struct T { ll b, c; }; // bx + c
18 inline S op(S a, S b) { return {
19     (a.x + b.x) % mod,
20     a.len + b.len,
21 }; }
22 inline S e() { return {0, 0}; }
23 inline S mapping(T f, S x) { return {
24     (f.b * x.x + f.c * x.len) % mod,
25     x.len,
26 }; }
27 inline T composition(T f, T g) { return {
28 // fb(gb*x + gc) + fc = (fb*gb)x + (fb*gc + fc)

```

```

29 (f.b * g.b) % mod,
30 (f.b * g.c + f.c) % mod,
31 }; }
32 inline T id() { return {1, 0}; }

```

4.8 Range Affine SegT

a2785d

```

1 #define mid ((l + r) >> 1)
2 #define cl (i<<1)
3 #define cr (i<<1|1)
4 class SegT {
5     int n; vector<S> a, t; vector<T> lz;
6     inline void pull(int i) { t[i] = op(t[cl], t[cr]); }
7     inline void apply(int i, T tag) {
8         t[i] = mapping(tag, t[i]);
9         lz[i] = composition(tag, lz[i]);
10    }
11    inline void push(int i) {
12        apply(cl, lz[i]); apply(cr, lz[i]);
13        lz[i] = id();
14    }
15    void build(int l, int r, int i) {
16        if (l == r) { t[i] = a[l]; return; } ///! a[l], not a[i]
17        build(l, mid, cl); build(mid+1, r, cr); pull(i);
18    }
19    int ql, qr; T tag;
20    void upd(int l, int r, int i) {
21        if (qr < l || r < ql) return;
22        if (ql <= l && r <= qr) { apply(i, tag); return; }
23        push(i); upd(l, mid, cl); upd(mid+1, r, cr); pull(i);
24    }
25    S qry(int l, int r, int i) {
26        if (qr < l || r < ql) return e();
27        if (ql <= l && r <= qr) return t[i];
28        push(i); return op(qry(l, mid, cl), qry(mid+1, r, cr));
29    }
30 public:
31     SegT(vector<S>& a):
32         n(a.size()), a(a), t(n<<2), lz(n<<2, id()) {
33         build(0, n - 1, 1);
34     }
35     void update(int l, int r, T tag) { // [l, r)
36         ql = l, qr = r - 1, this->tag = tag;
37         upd(0, n - 1, 1);
38     }
39     S query(int l, int r) { // [l, r)
40         ql = l, qr = r - 1;
41         return qry(0, n - 1, 1);
42     };
43 #undef mid
44 #undef cl
45 #undef cr

```

4.9 Range Affine SegT: Binary Search

658c18

```

1 // Add-on: paste inside SegT (before #undef mid/cl/cr).
2 // g: bool(S), monotone along the range, g(e()) must be true.
3 // e.g. first r with sum[l, r) >= k:
4 // max_right(l, [&](S s) { return s.x < k; }) (n if none)
5 // private:
6 template <class G> int mr(int l, int r, int i, S& acc, G g) {
7     if (r < ql) return -1;
8     if (ql <= l) { S s = op(acc, t[i]);
9         if (g(s)) { acc = s; return -1; }
10        if (l == r) return l; }
11    push(i);
12    int res = mr(l, mid, cl, acc, g);
13    return res != -1 ? res : mr(mid+1, r, cr, acc, g);
14 }
15 template <class G> int ml(int l, int r, int i, S& acc, G g) {
16     if (qr < l) return -1;
17     if (r <= qr) { S s = op(t[i], acc);
18         if (g(s)) { acc = s; return -1; }
19        if (l == r) return l + 1; }
20    push(i);
21    int res = ml(mid+1, r, cr, acc, g);
22    return res != -1 ? res : ml(l, mid, cl, acc, g);
23 }
24 // public:
25 // max r in [l, n] s.t. g(op(a[l..r))) true
26 template <class G> int max_right(int l, G g) {
27     ql = l; S acc = e(); int r = mr(0, n - 1, 1, acc, g);
28     return r == -1 ? n : r;
29 }
30 // min l in [0, r] s.t. g(op(a[l..r))) true
31 template <class G> int min_left(int r, G g) {
32     qr = r - 1; S acc = e(); int l = ml(0, n - 1, 1, acc, g);
33     return l == -1 ? 0 : l;

```


34 }

5 DP

5.1 SOS DP

d1c8c0

```
1// Author: Gino
2// Function: Solve problems that enumerates subsets of
  subsets (3^n => n*2^n)
3for (int msk = 0; msk < (1<<n); msk++) {
4    for (int i = 1; i <= n; i++) {
5        if (msk & (1<<(i - 1))) {
6            dp[msk][i] = dp[msk][i - 1] + dp[msk ^ (1<<(i - 1))][i - 1];
7        } else {
8            dp[msk][i] = dp[msk][i - 1];
9        }
10    }
11}
```

5.2 Divide and Conquer DP

7f9218

```
1/* Author: Simon Lindholm (License: CC0)
2 * Description: Given $a[i] = \min_{lo(i) \le k < hi(i)} (f(i, k))$ where the (minimal)
3 * optimal $k$ increases with $i$, computes $a[i]$ for $i = L..R-1$.
4 * Time: $O((N + (hi-lo)) \log N)$ */
5struct DP { // Modify at will:
6    int lo(int ind) { return 0; }
7    int hi(int ind) { return ind; }
8    ll f(int ind, int k) { return dp[ind][k]; }
9    void store(int ind, int k, ll v) { res[ind] = pii(k, v); }
10    void rec(int L, int R, int LO, int HI) {
11        if (L >= R) return;
12        int mid = (L + R) >> 1;
13        pair<ll, int> best(LLONG_MAX, LO);
14        rep(k, max(LO, lo(mid)), min(HI, hi(mid)))
15            best = min(best, make_pair(f(mid, k), k));
16        store(mid, best.second, best.first);
17        rec(L, mid, LO, best.second+1);
18        rec(mid+1, R, best.second, HI);
19    }
20    void solve(int L, int R) { rec(L, R, INT_MIN, INT_MAX); }
21};
```

5.3 Dynamic DP

- 適用情境：\$dp_i = M_i \cdot dp_{i-1} \Rightarrow dp_i = M_i M_{i-1} \dots M_1 dp_0\$
- 當 \$M_i\$ 需要動態修改，且 \$M_i\$ 是廣義矩陣乘法：

$$C_{ij} = \bigoplus_k A_{ik} \otimes B_{kj}$$

- 滿足 \$(\oplus, \otimes)\$ 是半環 (\$\otimes\$ 對 \$\oplus\$ 有分配律、\$\otimes\$ 和 \$\oplus\$ 有結合律、\$\oplus\$ 有交換律)
- 此時便可用線段樹維護 \$M_n M_{n-1} \dots M_1\$ 的乘積。線段樹上的 pos 是 \$M_{pos}\$，但 pull 時要 \$st[idx] = \text{mul}(st[cr], st[cl])\$

5.4 Convex Hull Trick

28a4d4

```
1/**
2 * Author: Simon Lindholm (License: CC0, Source: own work)
3 * Container where you can add lines of the form $kx+m$,
4 * and query maximum values at points $x$.
5 * Useful for dynamic programming ('`convex hull trick`').
6 * Time: $O(\log N)$, Status: stress-tested */
7struct Line {
8    mutable ll k, m, p;
9    bool operator<(const Line& o) const { return k < o.k; }
10    bool operator<(ll x) const { return p < x; }
11};
12struct LineContainer : multiset<Line, less<>> {
13    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
14    static const ll inf = LLONG_MAX;
15    ll div(ll a, ll b) { // floored division
16        return a / b - ((a ^ b) < 0 && a % b); }
17    bool isect(iterator x, iterator y) {
18        if (y == end()) return x->p = inf, 0;
19        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
20        else x->p = div(y->m - x->m, x->k - y->k);
21        return x->p >= y->p;
22    }
23    void add(ll k, ll m) {
24        auto z = insert({k, m, 0}); y = z++, x = y;
25        while (isect(y, z)) z = erase(z);
26        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
27        while ((y = x) != begin() && (--x)->p >= y->p)
28            isect(x, erase(y));
29    }
30    ll query(ll x) {
31        auto l = *lower_bound(x);
32        return l.k * x + l.m;
33    }
34};
```

6 Graph

6.1 Max Clique

3ea3ee

```
1// max N = 64
2typedef unsigned long long ll;
3struct MaxClique{
4    ll nb[ N ], n, ans;
5    void init( ll _n ){
6        n = _n;
7        for( int i = 0 ; i < n ; i ++ ) nb[ i ] = 0LLU;
8    }
9    void add_edge( ll _u , ll _v ){
10        nb[ _u ] |= ( 1LLU << _v );
11        nb[ _v ] |= ( 1LLU << _u );
12    }
13    void B( ll r , ll p , ll x , ll cnt , ll res ){
14        if( cnt + res < ans ) return;
15        if( p == 0LLU && x == 0LLU ){
16            if( cnt > ans ) ans = cnt;
17            return;
18        }
19        ll y = p | x; y &= -y;
20        ll q = p & ( ~nb[ int( log2( y ) ) ] );
21        while( q ){
22            ll i = int( log2( q & (-q) ) );
23            B( r | ( 1LLU << i ) , p & nb[ i ] , x & nb[ i ] ,
24              cnt + 1LLU , __builtin_popcountll( p &
25              nb[ i ] ) );
26            q &= ~( 1LLU << i );
27            p &= ~( 1LLU << i );
28            x |= ( 1LLU << i );
29        }
30    }
31    int solve(){
32        ans = 0;
33        ll _set = 0;
34        if( n < 64 ) _set = ( 1LLU << n ) - 1;
35        else{
36            for( ll i = 0 ; i < n ; i ++ ) _set |= ( 1LLU << i );
37        }
38        B( 0LLU , _set , 0LLU , 0LLU , n );
39        return ans;
40    }
41};
```

6.2 Bellman-Ford

841372

```
1// Author: Gino
2// n, m = #(vertices), #(edges), and vertices are 1-based
3// 1. BellmanFord bf(G, n, m);
4// 2-1: bf.calcDis(s);
5// => bf.dis[u] := shortest dis from s to u
6// => bf.dis[u] := LINF (s can't reach u)
7// => bf.dis[u] := -LINF (dis[u] can be arbitrary small)
8// 2-2: bf.findNegCycle();
9// => false (if no neg cycle)
10// => true (has neg cycle & gives an example: bf.neg_cycle)
11// Time: $O(VE)$, Space: $O(V + E)$
12const ll LINF = 4e18;
13struct BellmanFord {
14    const vector<vector<pair<int, ll>>& G;
15    int n, m; // #(vertices), #(edges)
16    BellmanFord(const auto& G, int n, int m):G(G), n(n), m(m) {}
17    vector<ll> dis; vector<int> nth_relax, pa;
18    void run_bf(const auto& src) {
19        dis.assign(n+1, LINF);
20        nth_relax.assign(n+1, 0); pa.assign(n+1, -1);
21        for (auto& s : src) dis[s] = 0;
22        for (int rlx = 1; rlx <= n; rlx++) {
23            for (int u = 1; u <= n; u++) {
24                if (dis[u] == LINF) continue; // !important
25                for (auto& [v, w] : G[u]) {
26                    if (dis[v] > dis[u] + w) {
27                        dis[v] = dis[u] + w; pa[v] = u;
28                        if (rlx == n) nth_relax[v] = 1;
29                    }
30                }
31            }
32        }
33        void calcDis(int s) {
34            run_bf(vector<int>{s}); queue<int> q;
35            for (int u=1; u<=n; u++) if (nth_relax[u]) q.push(u);
36            while (!q.empty()) { int u = q.front(); q.pop();
37                for (auto& [v, w] : G[u]) {
38                    if (!nth_relax[v]) {
39                        nth_relax[v] = 1; q.push(v);
40                    }
41                }
42            }
43            for (int u=1; u<=n; u++) if (nth_relax[u]) dis[u] = -LINF;
44        }
45        vector<int> neg_cycle;
46        bool findNegCycle() {
47            auto src = views::iota(1, n + 1); run_bf(src);
48            auto it = ranges::find_if(src,
```

```

44     [&](int s){ return nth_relax[s]; });
45     if (it == src.end()) return false;
46     int ptr = *it; for (int i=0; i<n; i++) ptr = pa[ptr];
47     neg_cycle.clear(); int cur = ptr;
48     while (true) {
49         neg_cycle.emplace_back(cur);
50         if (cur == ptr && neg_cycle.size() > 1) break;
51         cur = pa[cur];
52     }
53     ranges::reverse(neg_cycle); return true;
54 } };

```

6.3 System of Difference Constraints

```

1 vector<vector<pair<int, ll>>> G;
2 void add(int u, int v, ll w) { G[u].push_back({v, w}); }

```

- $x_u - x_v \leq c \Rightarrow \text{add}(v, u, c)$
- $x_u - x_v \geq c \Rightarrow \text{add}(u, v, -c)$
- $x_u - x_v = c \Rightarrow \text{add}(v, u, c), \text{add}(u, v, -c)$
- $x_u \geq c \Rightarrow \text{add super vertex } x_0 = 0, \text{ then } x_u - x_0 \geq c \Rightarrow \text{add}(u, 0, -c)$
- Don't forget non-negative constraints for every variable if specified implicitly.
- Interval sum $\Rightarrow$ Use prefix sum to transform into differential constraints. Don't forget $S_{i+1} - S_i \geq 0$ if x_i needs to be non-negative.
- $x_u/x_v \leq c \Rightarrow \log x_u - \log x_v \leq \log c$

6.4 Graph Girth

Run BFS for every node, when encountered non-BFS-tree edge, update answer (min cycle length) with $\text{dis}[u] + \text{dis}[v] + 1$. Time $O(VE)$.

6.5 Euler Trail

ee23d4

```

1 // Author: kactl (Simon Lindholm), wrapped by Gino
2 // Vertices are 1-based, Edge ID are 0-based
3 // 1. EulerWalk euler(n, true(directed) / false(undirected));
4 // 2. euler.addEdge(u, v);
5 // 3. int result = euler.solve();
6 // => -1: nothing (euler.path, euler.path_edges = {})
7 // => 1: Euler trail,
8 // => 2: Euler circuit
9 //     euler.path = {u_1, ..., u_{m+1}}
10 //     euler.path_edges = {e_1, ..., e_m}
11 // Time:  $O(V + E)$ , Space:  $O(V + E)$ 
12 // Test:  $V, E \leq 2e5, 99ms$ 
13 struct EulerWalk {
14     int n, m = 0; bool dir;
15     vector<vector<pii>> G; vi deg;
16     EulerWalk(int n, bool dir) : n(n), dir(dir), G(n+1), deg(n+1) {}
17     void addEdge(int u, int v) {
18         G[u].push_back({v, m});
19         if (!dir) G[v].push_back({u, m});
20         deg[u]++, deg[v] += dir ? -1 : 1; m++;
21     }
22     vi path, path_edges;
23     void walk(int src) {
24         path.clear(); path_edges.clear();
25         vi its(n+1), visE(m);
26         function<void(int, int)> dfs = [&](int u, int e_in) {
27             while (its[u] < sz(G[u])) {
28                 auto [v, e] = G[u][its[u]++];
29                 if (visE[e]) continue;
30                 visE[e] = 1;
31                 dfs(v, e);
32             }
33             path.push_back(u);
34             if (e_in != -1) path_edges.push_back(e_in);
35         };
36         dfs(src, -1);
37         if (sz(path) != m+1) {
38             path.clear(); path_edges.clear(); return;
39         }
40         ranges::reverse(path); ranges::reverse(path_edges);
41     }
42     int solve() {
43         int s = -1, odd = 0, src = 0, dst = 0;
44         for (int u = 1; u <= n; u++) {
45             if (!dir && (deg[u] & 1)) odd++, s = u;
46             if (dir && deg[u] == 1) s = u, src++;
47             if (dir && deg[u] == -1) dst++;
48             if (dir && abs(deg[u]) > 1) return -1;
49         }
50         if (!dir && odd != 0 && odd != 2) return -1;
51         if (dir && !(src == 0 && dst == 0) || (src == 1 && dst == 1)) return -1;
52         if (s == -1) for (int u = 1; u <= n; u++)
53             if (!G[u].empty()) { s = u; break; }
54         if (s == -1) s = 1;

```

```

57     walk(s);
58     return path.empty() ? -1 :
59         path.front() == path.back() ? 2 : 1;
60 } };

```

6.6 Vertex BCC (Round Square Tree)

21bcde

```

1 // Author: Ian, https://judge.yosupo.jp/submission/399308
2 // Vertex BCC (Round Square Tree) 0-based; Edge ID 0-based
3 // 1. BCC bcc(n); // n = #(vertices)
4 // 2. bcc.add_edge(u, v);
5 // 3. bcc.build();
6 // => bcc[id]: vertices in bcc `id` (square node n+id in T)
7 // => T[id]: tree adjacency; square nodes are n..n+ncnt-1
8 // => is_cut(x): whether x is a cut vertex
9 // => bcc_id(u, v): square node containing tree-edge (u, v)
10 struct BCC {
11     int n, m, t, ncnt;
12     vector<vector<pii>> G;
13     vector<vector<int>> T, bcc;
14     vector<int> dep, low, pa2, dep2, s;
15     void add_edge(int u, int v) {
16         G[u].eb(v, m), G[v].eb(u, m++); }
17     void dfs(int x, int f) {
18         if (~dep[x]) return;
19         dep[x] = low[x] = t++;
20         s.push_back(x);
21         for (auto [y, e]: G[x]) if (e != f) {
22             if (dep[y] == -1) {
23                 dfs(y, e), low[x] = min(low[x], low[y]);
24                 if (low[y] >= dep[x]) {
25                     int id = n + ncnt++, p;
26                     bcc.resize(bcc.size()+1);
27                     do {
28                         p = s.back(), s.pop_back();
29                         T[id].push_back(p), T[p].push_back(id);
30                         bcc.back().push_back(p);
31                     } while (p != y);
32                     T[id].push_back(x), T[x].push_back(id);
33                     bcc.back().push_back(x);
34                 }
35             } else low[x] = min(low[x], dep[y]); } }
36     void dfs2(int x, int p) {
37         dep2[x] = ~p ? dep2[p] + 1 : 0, pa2[x] = p;
38         for (auto y: T[x]) if (y != p) dfs2(y, x); }
39     bool is_cut(int x) { return (int)T[x].size() != 1; }
40     int bcc_id(int u, int v) {
41         return pa2[dep2[u] < dep2[v] ? v : u] - n; }
42     void build() {
43         for (int i = 0; i < n; i++) if (dep[i] == -1) {
44             if (G[i].empty()) { // isolated vertex: singleton bcc
45                 dep[i] = low[i] = t++;
46                 int id = n + ncnt++;
47                 bcc.push_back({i});
48                 T[id].push_back(i), T[i].push_back(id);
49             } else dfs(i, -1);
50             dfs2(i, -1);
51         } }
52     BCC(int n): n(n), m(0), t(0), ncnt(0), G(n), T(2*n),
53         dep(n, -1), low(n), pa2(2*n), dep2(2*n) {}
54 };

```

6.7 Edge BCC

e56b23

```

1 // Author: Ian, https://judge.yosupo.jp/submission/399295
2 // Edge BCC, 0-based; Edge ID 0-based
3 // 1. EBCC ebcc(n); // n = #(vertices)
4 // 2. ebcc.add_edge(u, v);
5 // 3. ebcc.build();
6 // => bcc[i]: each edge bcc
7 // => bccn[x]: in which bcc is x
8 // => isbr[e]: whether edge e is a bridge
9 struct EBCC {
10     int n, m, t, ncnt;
11     vector<vector<pii>> G;
12     vector<vector<int>> bcc;
13     vector<bool> isbr;
14     vector<int> dep, low, bccn, s;
15     EBCC(int n): n(n), m(0), t(0), ncnt(0), G(n),
16         dep(n, -1), low(n), bccn(n) {}
17     void add_edge(int u, int v) {
18         G[u].eb(v, m), G[v].eb(u, m++); }
19     void dfs(int x, int f) {
20         if (~dep[x]) return;
21         dep[x] = low[x] = t++;
22         s.push_back(x);
23         for (auto [y, e]: G[x]) if (e != f) {
24             if (dep[y] == -1)
25                 dfs(y, e), low[x] = min(low[x], low[y]);
26             else
27                 low[x] = min(low[x], dep[y]);

```

```

28     }
29     if (dep[x] == low[x]) {
30         if (~f) isbr[f] = true;
31         bcc.resize(bcc.size()+1);
32         int p; do {
33             bccn[p] = s.back() = ncnt;
34             s.pop_back();
35             bcc.back().push_back(p);
36         } while (p != x); ncnt++;
37     } }
38     void build() {
39         isbr.assign(m, false);
40         for (int i=0; i<n; i++) dfs(i, -1); }
41 };

```

6.8 Kth Shortest Path

15319c

```

1 // time: O(|E| lg |E|+|V| lg |V|+K)
2 // memory: O(|E| lg |E|+|V|)
3 struct KSP { // 1-base
4     struct nd{
5         int u,v; ll d;
6         nd(int ui=0,int vi=0,ll di=INF){ u=ui; v=vi; d=di; }
7     };
8     struct heap{ nd* edge; int dep; heap* chd[4]; };
9     static int cmp(heap* a, heap* b)
10     { return a->edge->d > b->edge->d; }
11     struct node{
12         int v; ll d; heap* H; nd* E;
13         node(ll _d,int _v,nd* _E){ d=_d; v=_v; E=_E; }
14         node(heap* _H,ll _d){ H=_H; d=_d; }
15         friend bool operator<(node a,node b)
16         { return a.d>b.d; }
17     };
18 };
19 int n,k,s,t,dst[N]; nd *nxt[N];
20 vector<nd*> g[N],rg[N]; heap *nullNd,*head[N];
21 void init(int _n,int _k,int _s,int _t){
22     n=_n; k=_k; s=_s; t=_t;
23     for(int i=1;i<=n;i++){
24         g[i].clear(); rg[i].clear();
25         nxt[i]=NULL; head[i]=NULL; dst[i]=-1;
26     }
27     void addEdge(int ui,int vi,ll di){
28         nd* e=new nd(ui,vi,di);
29         g[ui].push_back(e); rg[vi].push_back(e);
30     }
31     queue<int> dfsQ;
32     void dijkstra(){
33         while(dfsQ.size()) dfsQ.pop();
34         priority_queue<node> Q; Q.push(node(0,t,NULL));
35         while (!Q.empty()){
36             node p=Q.top(); Q.pop(); if(dst[p.v]!=-1)continue;
37             dst[p.v]=p.d; nxt[p.v]=p.E; dfsQ.push(p.v);
38             for(auto e:rg[p.v]) Q.push(node(p.d+e->d,e->u,e));
39         }
40         heap* merge(heap* curNd,heap* newNd){
41             if(curNd==nullNd) return newNd;
42             heap* root=new heap;memcpy(root,curNd,sizeof(heap));
43             if(newNd->edge->d<curNd->edge->d){
44                 root->edge=newNd->edge;
45                 root->chd[2]=newNd->chd[2];
46                 root->chd[3]=newNd->chd[3];
47                 newNd->edge=curNd->edge;
48                 newNd->chd[2]=curNd->chd[2];
49                 newNd->chd[3]=curNd->chd[3];
50             }
51             if(root->chd[0]->dep<root->chd[1]->dep)
52                 root->chd[0]=merge(root->chd[0],newNd);
53             else root->chd[1]=merge(root->chd[1],newNd);
54             root->dep=max(root->chd[0]->dep,
55                         root->chd[1]->dep)+1;
56             return root;
57         }
58     vector<heap*> V;
59     void build(){
60         nullNd=new heap; nullNd->dep=0; nullNd->edge=new nd;
61         fill(nullNd->chd,nullNd->chd+4,nullNd);
62         while(not dfsQ.empty()){
63             int u=dfsQ.front(); dfsQ.pop();
64             if(!nxt[u]) head[u]=nullNd;
65             else head[u]=head[nxt[u]->v];
66             V.clear();
67             for(auto&& e:g[u]){
68                 int v=e->v;
69                 if(dst[v]==-1) continue;
70                 e->d+=dst[v]-dst[u];
71                 if(nxt[u]!=e){
72                     heap* p=new heap;fill(p->chd,p->chd+4,nullNd);
73                     p->dep=1; p->edge=e; V.push_back(p);

```

```

74     }
75     }
76     if(V.empty()) continue;
77     make_heap(V.begin(),V.end(),cmp);
78     #define L(X) ((X<<1)+1)
79     #define R(X) ((X<<1)+2)
80     for(size_t i=0;i<V.size();i++){
81         if(L(i)<V.size()) V[i]->chd[2]=V[L(i)];
82         else V[i]->chd[2]=nullNd;
83         if(R(i)<V.size()) V[i]->chd[3]=V[R(i)];
84         else V[i]->chd[3]=nullNd;
85     }
86     head[u]=merge(head[u],V.front());
87 } }
88 vector<ll> ans;
89 void first_K(){
90     ans.clear(); priority_queue<node> Q;
91     if(dst[s]==-1) return;
92     ans.push_back(dst[s]);
93     if(head[s]!=nullNd)
94         Q.push(node(head[s],dst[s]+head[s]->edge->d));
95     for(int _=1; _<k and not Q.empty(); _++){
96         node p=Q.top(); q=Q.pop(); ans.push_back(p.d);
97         if(head[p.H->edge->v]!=nullNd){
98             q.H=head[p.H->edge->v]; q.d=p.d+q.H->edge->d;
99             Q.push(q);
100         }
101         for(int i=0;i<4;i++){
102             if(p.H->chd[i]!=nullNd){
103                 q.H=p.H->chd[i];
104                 q.d=p.d-p.H->edge->d+p.H->chd[i]->edge->d;
105                 Q.push(q);
106             }
107         }
108     void solve(){ // ans[i] stores the i-th shortest path
109         dijkstra(); build();
110         first_K(); // ans.size() might less than k
111     }
112 } solver;

```

6.9 SCC - Tarjan

56df66

```

1 // Author: Ian, https://judge.yosupo.jp/submission/399291
2 // Tarjan SCC, 0-based
3 // 1. SCC scc(n); // n = #(vertices)
4 // 2. Fill in adj list G
5 // 3. scc.build()
6 // => scc[i]: each scc
7 // => sccn[x]: in which scc is x
8 struct SCC {
9     const int inf = 1e9+5;
10     int n, t, ncnt;
11     vector<vector<int>> G, scc;
12     vector<bool> ins;
13     vector<int> sccn, dfn, low, s;
14     SCC(int n): n(n), t(0), ncnt(0), G(n),
15         ins(n, false), sccn(n), dfn(n, -1), low(n, inf) {}
16     void dfs(int x) {
17         if (~dfn[x]) return;
18         dfn[x] = low[x] = t++;
19         s.push_back(x), ins[x] = true;
20         for (auto i: G[x])
21             if (dfs(i), ins[i])
22                 low[x] = min(low[x], low[i]);
23         if (dfn[x] == low[x]) {
24             scc.resize(scc.size()+1);
25             int p; do {
26                 ins[p] = s.back() = false;
27                 s.pop_back(); sccn[p] = ncnt;
28                 scc.back().push_back(p);
29             } while (p != x); ncnt++;
30         }
31     }
32     void build() { for (int i=0; i<n; i++) dfs(i); }
33 };

```

6.10 2SAT

8d59f5

```

1 // Author: Ian, CryptoGraper
2 struct TwoSAT: SCC {
3     TwoSAT(int n): SCC(2*n) {}
4     void add_disjunction(int a,int na,int b,int nb) {
5         a = 2*a^na, b = 2*b^nb;
6         G[a^1].push_back(b);
7         G[b^1].push_back(a);
8     }
9     vector<int> solve() {
10         build(); vector<int> assignment(n/2,0);
11         for(int i=0; i<n/2; i++) {
12             if (sccn[2*i] == sccn[2*i^1]) return {};
13             assignment[i] = sccn[2*i] > sccn[2*i^1];
14         } return assignment;
15     };

```

```

16 // a -> b: (a,0,b,1)    not both a, b: (a,0,b,0)
17 // a and b: (a,1,a,1), (b,1,b,1)
18 // a or b: (a,1,b,1)    a xor b: (a,1,b,1), (a,0,b,0)
19 // a == b: (a,0,b,1), (a,1,b,0)
20 // force a = v: (a,v,a,v)
21 // solve(): assignment[i] = value of var i, {} if invalid

```

7 Tree

7.1 Tree Isomorphism (Rooted Trees)

a615bc

```

1 // Author: Gino
2 // Checks if 2 rooted trees are isomorphic
3 // Time: O(V log V)
4 // vector<vector<int>> G1, G2;
5 // int root_1, root_2;
6 map<vector<int>, int> dict;
7 int encode(const auto& G, int u, int pa = -1) {
8     vector<int> codes;
9     for (auto& v : G[u])
10         if (v != pa)
11             codes.emplace_back(encode(G, v, u));
12     ranges::sort(codes);
13     auto [it, _] = dict.try_emplace(codes, dict.size());
14     return it->second;
15 }
16 bool isomorphic() {
17     dict.clear(); dict[{}] = 0;
18     return encode(G1, root_1) == encode(G2, root_2);
19 }

```

7.2 Tree Isomorphism (Unrooted Trees)

Find the centroid(s) of T_1, T_2 .

Case 1: T_1, T_2 have different number of centroids $\rightarrow$ NO

Case 2: T_1 has centroid c_1, T_2 has centroid c_2

$\rightarrow$ rooted_isomorphic(c_1, c_2)

Case 3: T_1 has centroids c_1, c_1', T_2 has centroids c_2, c_2'

$\rightarrow$ rooted_isomorphic(c_1, c_2) || rooted_isomorphic(c_1', c_2')

7.3 Heavy Light Decomposition

16d094

```

1 // Author: Ian, https://judge.yosupo.jp/submission/399317
2 void segbuild(vector<int>&v);
3 void segmodify(int p, int k);
4 int segquery(int ql, int qr);
5 // Insert [ql, qr] segment tree here
6 template<class T=int> struct HLD {
7     int n, cnt;
8     vector<int> d, f, sz, son, top, dfn;
9     HLD(int n): n(n), cnt(0), d(n, 0), f(n, 0), sz(n, 1),
10         son(n, -1), top(n, 0), dfn(n, -1) {}
11     void dfs1(int x, int p) {
12         for (auto i: e[x]) if (i != p) {
13             d[i] = d[x]+1, f[i] = x;
14             dfs1(i, x), sz[x] += sz[i];
15             if (son[x] == -1 || sz[son[x]] < sz[i])
16                 son[x] = i;
17         }
18     }
19     void dfs2(int x, int t) {
20         dfn[x] = cnt++, top[x] = t;
21         if (son[x] == -1) return;
22         dfs2(son[x], t);
23         for (auto i: e[x])
24             if (!dfn[i]) dfs2(i, i);
25     }
26     void build(const vector<T> &v) {
27         dfs1(0, 0); dfs2(0, 0);
28         vector<T> dfnv(v.size());
29         for (int i=0; i<n; i++)
30             dfnv[dfn[i]] = v[i];
31         segbuild(dfnv);
32     }
33     void modify(int p, T v) { segmodify(dfn[p], v); }
34     T query(int a, int b) {
35         auto aggr = [](int x, int y) { return max(x, y); };
36         T ret = 0;
37         while (top[a] != top[b]) {
38             if (d[top[a]] > d[top[b]]) swap(a, b);
39             ret = aggr(ret, segquery(dfn[top[b]], dfn[b]+1));
40             b = f[top[b]];
41         }
42         if (dfn[a] > dfn[b]) swap(a, b);
43         return aggr(ret, segquery(dfn[a], dfn[b]+1));
44     }
45 };

```

7.4 Virtual Tree

5a0bd4

```

1 // Author: std_abs
2 // need lca, in, out
3 vector<pii> virtual_tree(vector<int> &v) {
4     auto cmp = [&](int x, int y) { return in[x] < in[y]; };
5     sort(all(v), cmp);
6     for (int i = 1; i < sz(v); ++i)

```

```

7     v.pb(lca(v[i-1], v[i]));
8     sort(all(v), cmp);
9     v.resize(unique(all(v)) - v.begin());
10    vector<int> stk(1, v[0]);
11    vector<pii> res;
12    for (int i = 1; i < sz(v); ++i) {
13        int x = v[i];
14        while (out[stk.back()] < out[x]) stk.pop_back();
15        res.emplace_back(stk.back(), x), stk.pb(x);
16    }
17    return res;
18 }

```

8 Matching

8.1 Bipartite Matching

e0a92e

```

1 // Author: Gino
2 // Function: Max Bipartite Matching in O(V sqrt(E))
3 //> hk.init(nx, ny) -> hk.add(x, y (+nx))
4 //> hk.max_matching(): the matching plan stores in mx, my
5 //> hk.min_vertex_cover(): vertex cover plan stores in vcover
6 // (!) vertices are 0-based: X = [0, nx), Y = [nx, nx+ny)
7 #define pb emplace_back
8 struct HopcroftKarp {
9     int n, nx, ny; vector<vector<int>> G; vector<int> mx, my;
10    HopcroftKarp(int nx, int ny): nx(nx), ny(ny), n(nx + ny) {
11        G.assign(n, vector<int>()); }
12    void add(int x, int y) { G[x].pb(y); G[y].pb(x); }
13    int max_matching() {
14        vector<int> dis, vis;
15        mx.assign(n, -1); my.assign(n, -1);
16        function<bool(int)> dfs = [&](int x) {
17            vis[x] = 1;
18            for (int y : G[x]) { int p = my[y];
19                if (p == -1 ||
20                    (dis[p] == dis[x] + 1 && !vis[p] && dfs(p)))
21                    return mx[x] = y, my[y] = x, true;
22            }
23            return false;
24        };
25        while (true) {
26            dis.assign(n, -1);
27            queue<int> q;
28            for (int x = 0; x < nx; x++)
29                if (mx[x] == -1) dis[x] = 0, q.push(x);
30            while (!q.empty()) {
31                int x = q.front(); q.pop();
32                for (int y : G[x])
33                    if (my[y] != -1 && dis[my[y]] == -1)
34                        dis[my[y]] = dis[x] + 1, q.push(my[y]);
35            }
36            vis.assign(n, 0);
37            bool found = false;
38            for (int x = 0; x < nx; x++)
39                if (mx[x] == -1 && dfs(x)) found = true;
40            if (!found) break;
41        }
42        int ans = 0;
43        for (int x = 0; x < nx; x++) if (mx[x] != -1) ans++;
44        return ans;
45    }
46    vector<int> vcover;
47    int min_vertex_cover() {
48        int ans = max_matching(); vector<int> vis(n, 0);
49        function<void(int)> dfs = [&](int x) {
50            vis[x] = true;
51            for (int y : G[x]) {
52                if (y == mx[x] || my[y] == -1 || vis[y]) continue;
53                vis[y] = true;
54                dfs(my[y]);
55            }
56        };
57        for (int x = 0; x < nx; x++) if (mx[x] == -1) dfs(x);
58        vcover.clear();
59        for (int x=0; x < nx; x++) if (!vis[x]) vcover.pb(x);
60        for (int y=nx; y < nx+ny; y++) if (vis[y]) vcover.pb(y);
61        return ans;
62    }
};

```

8.2 Bipartite Weighted Matching

5b976c

```

1 // Author: ckiseki, n = #(left vertices) = #(right vertices)
2 // left, right vertices labeled 1 ~ n separately
3 // 1. build l-based matrix G of size (n + 1) * (n + 1)
4 // G[u][v] := weight between (left u, right v)
5 // <!-- if not need perfect matching
6 // ==> set weights of all negative edges to 0
7 // 2. construct (solve at same time): KM km(n, G);
8 // ==> km.ans (max perfect weight sum)
9 // ==> km.fl[u] (left u -> right v)
10 // ==> km.fr[v] (right v -> left u)
11 // Time: O(V^3), Space: O(V^2), Test: V <= 500, 243ms

```



```

12 struct KM { // maximize, test @ UOJ 80
13     int n, l, r; ll ans; // fl and fr are the match
14     vector<ll> hl, hr; vector<int> fl, fr, pre, q;
15     void bfs(const auto &w, int s) {
16         vector<int> vl(n+1), vr(n+1);
17         vector<ll> slk(n+1, LINF);
18         l = r = 0; vr[q[r++]] = s; s = true;
19         auto check = [&](int x) -> bool {
20             if (vl[x] || slk[x] > 0) return true;
21             vl[x] = true; slk[x] = LINF;
22             if (fl[x] != -1) return (vr[q[r++]] = fl[x]) == true;
23             while (x != -1) swap(x, fr[fl[x] = pre[x]]);
24             return false;
25         };
26         while (true) {
27             while (l < r)
28                 for (int x = l, y = q[l++]; x <= n; ++x) if (!vl[x])
29                     if (ll val = hl[x] + hr[y] - w[x][y]; val < slk[x]) {
30                         slk[x] = val;
31                         if (pre[x] = y, !check(x)) return;
32                     }
33             ll d = LINF;
34             for (int x = l; x <= n; ++x) d = min(d, slk[x]);
35             for (int x = l; x <= n; ++x)
36                 vl[x] ? hl[x] += d : slk[x] -= d;
37             for (int x = l; x <= n; ++x) if (vr[x]) hr[x] -= d;
38             for (int x = l; x <= n; ++x) if (!check(x)) return;
39         }
40     }
41     KM(int n_, const auto &w) : n(n_), ans(0),
42         hl(n+1), hr(n+1), fl(n+1, -1), fr(fl), pre(n+1), q(n+1) {
43         for (int i = 1; i <= n; ++i) {
44             hl[i] = -LINF;
45             for (int j = 1; j <= n; ++j) hl[i] = max(hl[i], (ll)w[i][j]);
46         }
47         for (int i = 1; i <= n; ++i) bfs(w, i);
48         for (int i = 1; i <= n; ++i) ans += w[i][fl[i]];
49     };

```

8.3 General Matching

6e783c

```

1 // Author: ckiseki
2 // 0. build 1-based graph vector<vector<int>> G(n+1);
3 // 1. construct (solve at same time):
4 //     GeneralMatching M(G, n); => M.ans
5 // status: M.match[x] (== -1 if not matched, 1 <= x <= n)
6 // Time: O(V^3), Space: O(V+E), Test: V <= 500, 11ms
7 struct GeneralMatching {
8     queue<int> q; int ans, n;
9     vector<int> fa, s, v, pre, match;
10    int Find(int u) {
11        return u == fa[u] ? u : fa[u] = Find(fa[u]);
12    }
13    int LCA(int x, int y) {
14        static int tk = 0; tk++; x = Find(x); y = Find(y);
15        for (; swap(x, y)) if (x != 0) {
16            if (v[x] == tk) return x;
17            v[x] = tk;
18            x = Find(pre[match[x]]);
19        }
20    }
21    void Blossom(int x, int y, int l) {
22        for (; Find(x) != l; x = pre[y]) {
23            pre[x] = y, y = match[x];
24            if (s[y] == 1) q.push(y), s[y] = 0;
25            for (int z : {x, y}) if (fa[z] == z) fa[z] = l;
26        }
27    }
28    bool Bfs(auto &g, int r) {
29        iota(all(fa), 0); ranges::fill(s, -1);
30        q = queue<int>(); q.push(r); s[r] = 0;
31        for (; !q.empty(); q.pop()) {
32            for (int x = q.front(); int u : g[x])
33                if (s[u] == -1) {
34                    if (pre[u] = x, s[u] = 1, match[u] == 0) {
35                        for (int a = u, b = x, last;
36                             b != 0; a = last, b = pre[a])
37                            last = match[b], match[b] = a, match[a] = b;
38                        return true;
39                    }
40                    q.push(match[u]); s[match[u]] = 0;
41                } else if (!s[u] && Find(u) != Find(x)) {
42                    int l = LCA(u, x);
43                    Blossom(x, u, l); Blossom(u, x, l);
44                }
45            }
46        }
47        return false;
48    }
49    GeneralMatching(auto &g, int n_) : ans(0), n(n_),
50        fa(n+1), s(n+1), v(n+1), pre(n+1, 0), match(n+1, 0) {
51        for (int x = 1; x <= n; ++x)
52            if (match[x] == 0) ans += Bfs(g, x);
53        for (int x = 1; x <= n; ++x)

```

```

50         if (match[x] == 0) match[x] = -1;
51     };

```

9 Flow

9.1 Flow Methods

9516b6

```

1 // Author: CryptoGrapheR (some modified by Gino)
2 Maximize  $c^T x$  subject to  $Ax \leq b, x \geq 0$ ;
3 with the corresponding symmetric dual problem,
4 Minimize  $b^T y$  subject to  $A^T y \geq c, y \geq 0$ .
5 Maximize  $c^T x$  subject to  $Ax \leq b$ ;
6 with the corresponding asymmetric dual problem,
7 Minimize  $b^T y$  subject to  $A^T y = c, y \geq 0$ .
8 Maximize  $\sum x$  subject to  $x_i + x_j \leq A_{ij}, x \geq 0$ ;
9 => Maximize  $\sum x$  subject to  $x_i + x_j \leq A_{ij}$ ;
10 => Minimize  $A^T y = \sum A_{ij} y_{ij}$  subject to for all  $v$ ,
11      $\sum_{i=v \text{ or } j=v} y_{ij} = 1, y_{ij} \geq 0$ 
12 => possible optimal solution:  $y_{ij} = \{0, 0.5, 1\}$ 
13 =>  $y' = 2y$ :  $\sum_{i=v \text{ or } j=v} y'_{ij} = 2, y'_{ij} = \{0, 1, 2\}$ 
14 => Minimum Bipartite perfect matching/2 ( $V_1=X, V_2=X, E=A$ )
15 General Graph:
16 |Max Ind. Set| + |Min Vertex Cover| = |V|
17 |Max Ind. Edge Set| + |Min Edge Cover| = |E|
18 Bipartite Graph:
19 |Max Ind. Set| = |Min Edge Cover|
20 |Max Ind. Edge Set| = |Min Vertex Cover|
21 To reconstruct the minimum vertex cover, dfs from each
22 unmatched vertex on the left side and with unused edges
23 only. Equivalently, dfs from source with unused edges
24 only and without visiting sink. Then, a vertex is
25 chosen iff. it is on the left side and without visited
26 or on the right side and visited through dfs.
27 Minimum Weighted Bipartite Edge Cover:
28 Construct new bipartite graph with  $n+m$  vertices on each
29 side:
30 for each vertex  $u$ , duplicate a vertex  $u'$  on the other side
31 for each edge  $(u,v,w)$ , add edges  $(u,v,w)$  and  $(v',u',w)$ 
32 for each vertex  $u$ , add edge  $(u,u',2w)$  where  $w$  is min edge
33 connects to  $u$ 
34 then the answer is the minimum perfect matching of the new
35 graph (KM)
36 Maximum density subgraph ( $\sum W_e + \sum W_v$ ) / |V|
37 Binary search on answer:
38 For a fixed  $D$ , construct a Max flow model as follow:
39 Let  $S$  be Sum of all weight (or inf)
40 1. from source to each node with cap =  $S$ 
41 2. For each  $(u,v,w)$  in  $E$ ,  $(u \rightarrow v, \text{cap}=w), (v \rightarrow u, \text{cap}=w)$ 
42 3. For each node  $v$ , from  $v$  to sink with cap =  $S + 2 * D - \text{deg}[v] - 2 * (W \text{ of } v)$ 
43 where  $\text{deg}[v] = \sum \text{weight of edge associated with } v$ 
44 If  $\text{maxflow} < S * |V|$ ,  $D$  is an answer.
45 Requiring subgraph: all vertex can be reached from source
46 with
47 edge whose cap > 0.
48 Maximum closed subgraph
49 1. connect source with positive weighted
50 vertex(capacity=weight)
51 2. connect sink with negative weighted vertex(capacity=-
52 weight)
53 3. make capacity of the original edges = inf
54 4. ans = sum(positive weighted vertex weight) - (max flow)
55 (Node-disjoint) Min DAG Path Cover (用最短路徑覆蓋所有點)
56 Node disjoint: 拆出來的路徑不能共用同一個點
57 將一個點  $u$  裂成  $u+$  和  $u-$ , 代表進入和出去
58 對於一條邊  $(u, v)$  建邊  $u- \Rightarrow v+$ 
59 1. 點們和 -點們形成一張二分圖
60 2. 最差的答案是  $n$ , 代表每個點自己一個點就是一條路徑
61 3. 只要一組  $(u-, v+)$  匹配成功那對應到的答案恰好會  $-1$ 
62 >>> ans =  $n - \text{這張二分圖的最大匹配}$ 
63 General DAG Path Cover
64 跟 Node-disjoint 版本差在點可以被很多條路共用
65 建邊方式 Node-disjoint 差別在條件比較鬆
66 Node-disjoint:  $u- \Rightarrow v+$  只能在  $(u, v)$  有邊時建
67 General:  $u- \Rightarrow v+$  在  $u$  能走到  $v$  的時候就建
68 Dilworth Theorem
69 反鏈：一些節點的集合，滿足這些節點互相無法抵達
70 最大反鏈 = 最小 General DAG Path Cover
71 Min Cut reconstruction (after running max flow  $s \rightarrow t$ ):
72 From  $s$ , BFS/DFS through residual edges with remaining
73 capacity > 0, marking reached vertices as visited.
74 The cut set is every original edge  $(u, v)$  with  $u$  visited
75 and  $v$  not visited; its capacity sum equals the max flow.
76 Bipartite Matching reconstruction (after running max flow
77 on source  $\rightarrow X \rightarrow Y \rightarrow \text{sink}$  model):
78 For each edge  $(x, y)$  between the  $X$ -side and  $Y$ -side, it is
79 part of the matching iff. its residual capacity is 0
80 (i.e. it is saturated).
81 Max/Min Flow with Lower Bound (Circulation Problem):
82 1. Construct super source  $S$  and super sink  $T$ .
83 2. For each edge  $(x, y, l, u)$ , connect  $x \rightarrow y$  with capacity
84  $u - l$ .

```

```

77 3. For each vertex  $v$ , let  $\text{in}(v) = (\text{sum of incoming lower bounds})$ 
78   - (sum of outgoing lower bounds).
79 4. If  $\text{in}(v) > 0$ , connect  $S \rightarrow v$  with capacity  $\text{in}(v)$ ;
80   otherwise connect  $v \rightarrow T$  with capacity  $-\text{in}(v)$ .
81 To maximize: connect  $t \rightarrow s$  with capacity  $\text{inf}$  (skip this
82 edge
83 for a pure circulation problem), let  $f = \text{maxflow}(S, T)$ .
84 If
85  $f \neq \text{sum of in}(v)$  over all  $\text{in}(v) > 0$ , no solution exists.
86 Otherwise  $\text{maxflow}(s, t)$  (found after adding  $t \rightarrow s$ ) is
87 the answer.
88 To minimize: let  $f = \text{maxflow}(S, T)$  (without any  $s-t$  edge
89 yet).
90 Then connect  $t \rightarrow s$  with capacity  $\text{inf}$  and let  $f'$  be the
91 extra
92 flow pushed from  $S$  to  $T$ . If  $f + f' \neq \text{sum of in}(v)$  over
93 all
94  $\text{in}(v) > 0$ , no solution exists. Otherwise  $f'$  is the
95 answer.
96 The flow on each original edge  $e$  is  $\text{l}_e + f_e$ , where  $f_e$ 
97 is
98  $e$ 's flow in the constructed graph.

```

9.2 Dinic (Flow Decomposition)

8d8937

```

1// Function: Max Flow + Flow Decomposition,  $O(V^2 E)$ 
2// init -> addEdge; vertices 0-based
3// >>> flow() := return max flow
4// >>> flow_decomposition() := return max flow + store
5// decomposition in flow.D
6struct Dinic{
7    static const int MXN=10000;
8    struct Edge{ int v,c,r,C; bool fw; };
9    int n,s,t,level[MXN]; vector<Edge> G[MXN];
10    void init(int _n,int _s,int _t){
11        n=_n; s=_s; t=_t;
12        for (int i=0; i<n; i++) G[i].clear();
13    }
14    void addEdge(int u,int v,int c){
15        G[u].push_back({v,c,(int)G[v].size(),c,true});
16        G[v].push_back({u,0,(int)G[u].size()-1,0,false});
17    }
18    bool BFS(){
19        for (int i=0; i<n; i++) level[i]=-1;
20        queue<int> que; que.push(s); level[s]=0;
21        while(!que.empty()){
22            int u=que.front(); que.pop();
23            for (auto &it:G[u]){
24                if(it.c>0&&level[it.v]==-1){
25                    level[it.v]=level[u]+1; que.push(it.v);
26                }
27            }
28        }
29        return level[t]!=-1;
30    }
31    int DFS(int u,int nf){
32        if(u==t) return nf;
33        int res=0;
34        for (auto &it:G[u]){
35            if(it.c>0&&level[it.v]==level[u]+1){
36                int tf=DFS(it.v,min(nf,it.c));
37                res+=tf; nf-=tf; it.c-=tf;
38                G[it.v][it.r].c+=tf;
39                if(nf==0) return res;
40            }
41        }
42        if(!res) level[u]=-1;
43        return res;
44    }
45    int flow(int res=0){
46        while(BFS()) res+=DFS(s,2147483647);
47        return res;
48    }
49    // reset: do nothing
50    vector<pair<int, vector<int>>> D; // (flow amount, [path
51    p1 ... pk])
52    int flow_decomposition(){
53        int mxflow = flow();
54        vector<vector<Edge>> fG(n); // graph with forward edges
55        for (int u = 0; u < n; u++) {
56            for (auto &e : G[u]) {
57                if (e.fw && e.c - e.c > 0)
58                    fG[u].push_back({e.v, e.c - e.c, e.r, e.C, e.fw});
59            }
60        }
61        vector<int> vis(n, 0);
62        function<int(int, int)> dfs = [&](int u, int cur) {
63            if (u == t) {
64                D.back().second.push_back(u);
65                return cur;
66            }
67            vis[u] = 1;
68            for (auto &e : fG[u]) {
69                if (e.c > 0 && !vis[e.v]) {
70                    int ans = dfs(e.v, min(cur, e.c));
71                    if (ans > 0) {
72                        e.c -= ans;
73                        D.back().second.push_back(u);
74                        return ans;
75                    }
76                }
77            }
78        };
79    }
80 }

```

```

64    return 0; };
65    D.clear();
66    int quota = mxflow;
67    while (quota > 0) {
68        D.emplace_back(0, vector<int>());
69        fill(vis.begin(), vis.end(), 0);
70        int f = dfs(s, 2147483647);
71        if (f == 0) break;
72        reverse(D.back().second.begin(),
73              D.back().second.end());
74        D.back().first = f, quota -= f;
75    }
76    return mxflow;
77 }

```

9.3 ISAP

647ff0

```

1// Author: CRYPTOGRAPHER
2// 1-based: vertices are numbered 1 ~ n
3// 1. flow.init(n, s, t);
4// n = #vertices, s, t = source, sink
5// 2. flow.addEdge(u, v, c); // u, v in [1, n]
6// 3. call: int ans = flow.flow();
7// => ans = max flow value from s to t
8// Time:  $O(V^2 * E)$  worst case, usually much faster in
9// practice
10static const int MAXV=50010;
11static const int INF =1000000;
12struct Maxflow{
13    struct Edge{
14        int v,c,r;
15        Edge(int _v,int _c,int _r):v(_v),c(_c),r(_r){}
16    };
17    int s,t; vector<Edge> G[MAXV];
18    int iter[MAXV],d[MAXV],gap[MAXV],tot;
19    void init(int n,int _s,int _t){
20        tot=n; s=_s; t=_t;
21        for(int i=1;i<=tot;i++){
22            G[i].clear(); iter[i]=d[i]=gap[i]=0;
23        }
24    }
25    void addEdge(int u,int v,int c){
26        G[u].push_back(Edge(v,c,SZ(G[v])));
27        G[v].push_back(Edge(u,0,SZ(G[u])-1));
28    }
29    int DFS(int p,int flow){
30        if(p==t) return flow;
31        for(int &i=iter[p]; i<SZ(G[p]); i++){
32            Edge &e=G[p][i];
33            if(e.c>0&&d[p]==d[e.v]+1){
34                int f=DFS(e.v,min(flow,e.c));
35                if(f){ e.c-=f; G[e.v][e.r].c+=f; return f; }
36            }
37        }
38        if(--gap[d[p]]==0) d[s]=tot;
39        else{ d[p]++; iter[p]=0; ++gap[d[p]]; }
40        return 0;
41    }
42    int flow(){
43        int res=0;
44        for(res=0; gap[0]=tot; d[s]<tot; res+=DFS(s,INF));
45        return res;
46    }
47    void reset(){
48        for(int i=1;i<=tot;i++) iter[i]=d[i]=gap[i]=0;
49    }
50 }

```

9.4 Bounded Max Flow

aaf309

```

1// Author: CRYPTOGRAPHER
2// Max flow with lower/upper bound on edges (source-sink
3// version)
4// <!=> 1-based: original vertices are numbered 1 ~ n.
5// Super source, sink: n+1, n+2 respectively.
6// <!=> dependency: ISAP.cpp
7// 1. n = #vertices (1 ~ n), m = #edges
8// s, t = original source, sink (both in [1, n])
9// 2. l[i], r[i] := edge i goes from l[i] -> r[i] (l[i],
10// r[i] in [1, n])
11// 3. a[i], b[i] := flow on edge i must lie in [a[i], b[i]]
12// 4. int ans = solve(n, m, s, t);
13// => ans == -1 : no feasible flow exists
14// => ans : feasible max flow value
15// <!=> N, M must be large enough for original graph + 2
16// super nodes
17// Time:  $O(V^2 E)$ , Space:  $O(V + E)$ 
18int in[N], out[N], l[M], r[M], a[M], b[M];
19int solve(int n, int m, int s, int t){
20    flow.init(n+2, n+1, n+2); // super source = n+1, super
21    sink = n+2
22    for(int i = 0; i < m; i++){

```

```

19   in[r[i]] += a[i]; out[l[i]] += a[i];
20   flow.addEdge(l[i], r[i], b[i] - a[i]);
21   // flow from l[i] to r[i] must lie in [a[i], b[i]]
22 }
23 int nd = 0;
24 for(int i = 1; i <= n; i++){
25   if(in[i] < out[i]){
26     flow.addEdge(i, flow.t, out[i] - in[i]);
27     nd += out[i] - in[i];
28   }
29   if(out[i] < in[i])
30     flow.addEdge(flow.s, i, in[i] - out[i]);
31 }
32 flow.addEdge(t, s, INF); // original sink -> source
33 if(flow.flow() != nd) return -1; // infeasible
34 int ans = flow.G[s].back().c; // t->s edge's current flow
= feasible flow s->t
35 flow.G[s].back().c = flow.G[t].back().c = 0;
36 for(size_t i = 0; i < flow.G[flow.s].size(); i++){
37   Maxflow::Edge &e = flow.G[flow.s][i];
38   flow.G[flow.s][i].c = 0; flow.G[e.v][e.r].c = 0;
39 }
40 for(size_t i = 0; i < flow.G[flow.t].size(); i++){
41   Maxflow::Edge &e = flow.G[flow.t][i];
42   flow.G[flow.t][i].c = 0; flow.G[e.v][e.r].c = 0;
43 }
44 flow.addEdge(flow.s, s, INF); flow.addEdge(t, flow.t,
INF);
45 flow.reset(); // keeps G[] intact, only resets search
state
46 return ans + flow.flow();
47 }

```

9.5 MCMF

b4675d

```

1// Author: CRYPTOGRAPHER
2// MCMF (supports negative edges without negative cycles)
3// <!-- 1-based: vertices are numbered 1 ~ n.
4// 1. MCMF.init(n); // n = #vertices (1 ~ n)
5// 2. MCMF.add(u, v, cap, cost);
6// 3. auto [max_flow, min_cost] = MCMF.flow(s, t); // s, t
in [1, n]
7// Time: O(F * E), Space: O(V + E)
8typedef int Tcost;
9const int MAXV = 20010;
10const int INF = 1000000;
11const Tcost INFC = 1e9;
12struct MCMF {
13  struct Edge{
14    int v, cap;
15    Tcost w;
16    int rev;
17    bool fw;
18    Edge(){}
19    Edge(int t2, int t3, Tcost t4, int t5, bool t6)
20      : v(t2), cap(t3), w(t4), rev(t5), fw(t6) {}
21  };
22  int V, s, t;
23  vector<Edge> G[MAXV];
24  void init(int n){
25    V = n;
26    for(int i = 0; i <= V; i++) G[i].clear();
27  }
28  void add(int a, int b, int cap, Tcost w){
29    G[a].push_back(Edge(b, cap, w, (int)G[b].size(), true));
30    G[b].push_back(Edge(a, 0, -w, (int)G[a].size()-1,
false));
31  }
32  Tcost d[MAXV];
33  int id[MAXV], mom[MAXV];
34  bool inqu[MAXV];
35  queue<int> q;
36  pair<int, Tcost> flow(int _s, int _t){
37    s = _s, t = _t;
38    int mxf = 0; Tcost mnc = 0;
39    while(1){
40      fill(d, d+1+V, INFC); // need to use type cast
41      fill(inqu, inqu+1+V, 0);
42      fill(mom, mom+1+V, -1);
43      mom[s] = s;
44      d[s] = 0;
45      q.push(s); inqu[s] = 1;
46      while(q.size()){
47        int u = q.front(); q.pop();
48        inqu[u] = 0;
49        for(int i = 0; i < (int) G[u].size(); i++){
50          Edge &e = G[u][i];
51          int v = e.v;
52          if(e.cap > 0 && d[v] > d[u]+e.w){
53            d[v] = d[u]+e.w;

```

```

54            mom[v] = u;
55            id[v] = i;
56            if(!inqu[v]) q.push(v), inqu[v] = 1;
57          }
58        }
59      }
60      if(mom[t] == -1) break;
61      int df = INF;
62      for(int u = t; u != s; u = mom[u])
63        df = min(df, G[mom[u]][id[u]].cap);
64      for(int u = t; u != s; u = mom[u]){
65        Edge &e = G[mom[u]][id[u]];
66        e.cap -= df;
67        G[e.v][e.rev].cap += df;
68      }
69      mxf += df;
70      mnc += df*d[t];
71    }
72    return make_pair(mxf, mnc);
73  }
74};

```

9.6 Push-Relabel

f50218

```

1// Author: Simon Lindholm (kactl)
2// 0-based: vertices are numbered 0 ~ n-1
3// 1. PushRelabel pr(n); // n = #vertices
4// 2. pr.addEdge(u, v, c); // u, v in [0, n-1]
5// 3. ll ans = pr.calc(s, t);
6// => ans = max flow value from s to t
7// To obtain actual flow:
8// for (int u = 0; u < n; u++){
9//   for (auto& e : pr.g[u])
10//     if (e.f > 0) => (u, e.dest, e.f)
11// Time: O(V^2 * sqrt(E)), better on dense graph
12struct PushRelabel {
13  struct Edge { int dest, back; ll f, c; };
14  vector<vector<Edge>> g;
15  vector<ll> ec;
16  vector<Edge*> cur;
17  vector<vi> hs; vi H;
18  PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n) {}
19  void addEdge(int s, int t, ll cap, ll rcap=0) {
20    if (s == t) return;
21    g[s].push_back({t, sz(g[t]), 0, cap});
22    g[t].push_back({s, sz(g[s])-1, 0, rcap});
23  }
24  void addFlow(Edge& e, ll f) {
25    Edge &back = g[e.dest][e.back];
26    if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
27    e.f += f; e.c -= f; ec[e.dest] += f;
28    back.f -= f; back.c += f; ec[back.dest] -= f;
29  }
30  ll calc(int s, int t) {
31    int v = sz(g); H[s] = v; ec[t] = 1;
32    vi co(2*v); co[0] = v-1;
33    rep(i, 0, v) cur[i] = g[i].data();
34    for (Edge& e : g[s]) addFlow(e, e.c);
35    for (int hi = 0;;) {
36      while (hs[hi].empty()) if (!hi--) return -ec[s];
37      int u = hs[hi].back(); hs[hi].pop_back();
38      while (ec[u] > 0) // discharge u
39        if (cur[u] == g[u].data() + sz(g[u])) {
40          H[u] = 1e9;
41          for (Edge& e : g[u]) if (e.c && H[u] >
H[e.dest]+1)
42            H[u] = H[e.dest]+1, cur[u] = &e;
43          if (++co[H[u]], !--co[hi] && hi < v)
44            rep(i, 0, v) if (hi < H[i] && H[i] < v)
45              --co[H[i]], H[i] = v + 1;
46          hi = H[u];
47        } else if (cur[u]->c && H[u] == H[cur[u]->dest]+1)
48          addFlow(*cur[u], min(ec[u], cur[u]->c));
49        else ++cur[u];
50      }
51    }
52    bool leftOfMinCut(int a) { return H[a] >= sz(g); }
53};

```

9.7 Gomory-Hu Tree

3777d4

```

1// Author: chilli, Takanori MAEHARA (kactl)
2// <!-- Dependency: PushRelabel.cpp
3// Given a list of edges representing an undirected flow
graph,
4// returns edges of the Gomory-Hu tree.
5// maxflow / mincut of (u, v) => min edge from u to v on
Gomory-Hu tree
6// Time: O(V) * O(PushRelabel)
7typedef array<ll, 3> Edge; // (u, v, w)
8vector<Edge> gomoryHu(int N, vector<Edge> ed) {
9  vector<Edge> tree;

```

```

10 vi par(N);
11 rep(i, 1, N) {
12     PushRelabel D(N); // Dinic also works
13     for (Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
14     tree.push_back({i, par[i], D.calc(i, par[i])});
15     rep(j, i+1, N)
16         if (par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
17 }
18 return tree;
19 }

```

9.8 Global Min Cut

a0d6c2

```

1 // Author: ckiseki
2 // 1-based: vertices are numbered 1 ~ n
3 // 1. vector<vector<ll>> G(n+1, vector<ll>(n+1, 0));
4 // 2. add_edge(G, u, v, c); // add undirected edge u-v with
   weight c, u, v in [1, n]
5 // 3. ll ans = mincut(G, n); // n = #vertices
6 // => ans = global min cut value
7 // Time: O(V^3)
8 void add_edge(auto &w, int u, int v, int c) {
9     w[u][v] += c; w[v][u] += c; }
10 auto phase(const auto &w, int n, vector<int> id) {
11     vector<ll> g(n+1); int s = -1, t = -1;
12     while (!id.empty()) {
13         int c = -1;
14         for (int i : id) if (c == -1 || g[i] > g[c]) c = i;
15         s = t; t = c;
16         id.erase(ranges::find(id, c));
17         for (int i : id) g[i] += w[c][i];
18     } return tuple{s, t, g[t]};
19 }
20 ll mincut(auto w, int n) {
21     ll cut = numeric_limits<ll>::max();
22     vector<int> id(n); iota(all(id), 1);
23     for (int i = 0; i < n - 1; ++i) {
24         auto [s, t, gt] = phase(w, n, id);
25         id.erase(ranges::find(id, t));
26         cut = min(cut, gt);
27         for (int j = 1; j <= n; ++j)
28             w[s][j] += w[t][j], w[j][s] += w[j][t];
29     } return cut;
30 }

```

10 String

10.1 Rolling Hash

bb1f6e

```

1 // Author: ckiseki (Modified by Gino)
2 // Usage: RH<Sigma, Mod> Hs(s);
3 // If no modifications needed, replace BIT with prefix sum
4 template <ll P = 127, ll Q = 1051762951>
5 class RH {
6     int n; vector<ll> p, inv, d, bit;
7     ll add(ll a, ll b) { return a+b >= Q ? a+b-Q : a+b; }
8     ll sub(ll a, ll b) { return a-b < 0 ? a-b+Q : a-b; }
9     ll mul(ll a, ll b) { return (__int128)a * b % Q; }
10    ll pw(ll a, ll x) {
11        ll r = 1; for (; x; a = mul(a, a), x >>= 1)
12            if (x & 1) r = mul(r, a);
13        return r;
14    }
15    void bit_add(int i, ll x)
16    { for (; i <= n; i += i&~i) bit[i] = add(bit[i], x); }
17    ll bit_gry(int i)
18    { ll s = 0; for (; i; i -= i&~i) s = add(s, bit[i]);
19    return s; }
20 public:
21     RH(const auto &s) : n(s.size()),
22         p(n+1, 1), inv(n+1, 1), d(n+1), bit(n+1) {
23         ll iv = pw(P, Q - 2);
24         for (int i = 1; i <= n; i++) {
25             p[i] = mul(p[i-1], P); inv[i] = mul(inv[i-1], iv);
26             d[i] = mul(s[i-1], p[i]); bit_add(i, d[i]);
27         }
28         void update(int i, char c)
29         { ll x = mul(c, p[i]); bit_add(i, sub(x, d[i])); d[i] = x; }
30         ll query(int l, int r)
31         { return mul(sub(bit_gry(r), bit_gry(l-1)), inv[l]); }
32     };

```

10.2 KMP

47ace9

```

1 // Author: Gino
2 vector<int> kmp(auto s) { int n=s.size(); vector<int> f(n,0);
3     for (int i = 1, ptr = 0; i < n; i++) {
4         while (ptr && s[i] != s[ptr]) ptr = f[ptr-1];
5         if (s[i] == s[ptr]) ptr++; f[i] = ptr;
6     } return f;
7 }

```

10.3 Z Value

20d446

```

1 // Author: std_abs
2 vector<int> Z(auto s) { int n=s.size(); vector<int> Z(n);

```

```

3     for (int i = 0, l = 0, r = 0; i < n; ++i) {
4         Z[i] = max(min(Z[i-l], r-i), 0);
5         while (i+Z[i] < n && s[Z[i]] == s[i+Z[i]])
6             l = i, r = i+Z[i], Z[i]++;
7     } return Z;
8 }

```

10.4 Manacher

846756

```

1 // Author: std_abs
2 // return array Z (length == 2N - 1)
3 // Example: s = "aaaa"
4 // a a a a
5 // Z_0 Z_1 Z_2 Z_3 Z_4 Z_5 Z_6
6 // 1 2 3 4 3 2 1
7 // The corresponding palindrome s[l..r]:
8 // l = (i - Z[i] + 1) / 2
9 // r = (i + Z[i] - 1) / 2
10 vector<int> manacher(auto tmp) {
11     vector<int> s(1, -1);
12     for (auto c : tmp) s.push_back(c), s.push_back(-1);
13     int l = 0, r = 0, n = (int)s.size();
14     vector<int> Z(n);
15     for (int i = 0; i < n; ++i) {
16         Z[i] = r > i ? min(Z[2 * l - i], r - i) : 1;
17         while (i - Z[i] >= 0 && i + Z[i] < n &&
18             s[i + Z[i]] == s[i - Z[i]]) Z[i]++;
19         if (Z[i] + i > r) l = i, r = Z[i] + i;
20     }
21     for (int i = 0; i < n; ++i)
22         Z[i] = (Z[i] - (i & 1)) / 2 * 2 + (i & 1);
23     return vector<int>(Z.begin() + 1, Z.end() - 1);
24 }

```

10.5 Suffix Array + LCP

15b646

```

1 // Author: std_abs
2 // Tested @yosupo, N <= 5e5, 123ms
3 const int N = 5e5 + 10;
4 int sa[N], tmp[2][N], c[N], rk[N], lcp[N];
5 void buildSA(string s) {
6     int *x = tmp[0], *y = tmp[1], m = 256, n = s.size();
7     for (int i = 0; i < m; ++i) c[i] = 0;
8     for (int i = 0; i < n; ++i) c[x[i]] = s[i]++;
9     for (int i = 1; i < m; ++i) c[i] += c[i-1];
10    for (int i = n-1; ~i; --i) sa[--c[x[i]]] = i;
11    for (int k = 1; k < n; k <= 1) {
12        for (int i = 0; i < m; ++i) c[i] = 0;
13        for (int i = 0; i < n; ++i) c[x[i]]++;
14        for (int i = 1; i < m; ++i) c[i] += c[i-1];
15        int p = 0;
16        for (int i = n-k; i < n; ++i) y[p++] = i;
17        for (int i = 0; i < n; ++i) if (sa[i] >= k)
18            y[p++] = sa[i] - k;
19        for (int i = n-1; ~i; --i)
20            sa[--c[x[y[i]]]] = y[i];
21        y[sa[0]] = p = 0;
22        for (int i = 1; i < n; ++i) {
23            int a = sa[i], b = sa[i-1];
24            p += !(x[a] == x[b] && a+k < n && b+k < n && x[a+k] == x[b+k]);
25            y[sa[i]] = p;
26        }
27        if (n == p+1) break;
28        swap(x, y), m = p+1;
29    }
30    void buildLCP(string s) {
31        // lcp[i] = LCP(sa[i-1], sa[i])
32        // lcp(i, j) = query_lcp_min[rk[i]+1, rk[j]+1]
33        int n = s.length(), x = 0;
34        for (int i = 0; i < n; ++i) rk[sa[i]] = i;
35        for (int i = 0; i < n; ++i) {
36            if (!rk[i]) lcp[rk[i]] = 0;
37            else { if (x) x--;
38                int p = sa[rk[i]-1];
39                while (x+i < n && x+p < n && s[x+i] == s[x+p]) x++;
40                lcp[rk[i]] = x;
41            } }

```

10.6 Suffix Automaton

322a41

```

1 // Author: std_abs
2 // state (i.e. node) -> all substr with the same endpos
3 // those substrs' length in range [len(link) + 1, len]
4 // state i's endpos set -> union of pos[i] over the subtree
   of i in the suffix link tree
5 // link[i] -> longest suffix with different endpos set
   (suffix link tree edge)
6 // len[i] -> length longest suffix of state i
7 // pos[i] -> an example end position of state i (1-based)
8 // cnt[i] -> size of endpos set of state i, built by
   build_cnt() (tree DP)
9 const int N = 5e5 + 10, C = 26;
10 struct SAM {

```



```

11 int ch[2*N][C], len[2*N], link[2*N], pos[2*N], cnt[2*N], _id;
12 SAM () { reset(); }
13 void reset() { _id = 0, newnode(), link[0] = -1; }
14 int newnode() {
15     fill_n(ch[_id], C, 0);
16     len[_id] = link[_id] = pos[_id] = cnt[_id] = 0;
17     return _id++;
18 }
19 void build(string s) {
20     int last = 0; for (int i = 0; i < (int)s.size(); i++) {
21         char c = s[i]; int cur = newnode();
22         len[cur] = len[last] + 1, pos[cur] = i + 1;
23         int p = last;
24         while (~p && !ch[p][c - 'a'])
25             ch[p][c - 'a'] = cur, p = link[p];
26         if (p == -1) link[cur] = 0;
27         else { // conflict: p --> q, p --> cur
28             int q = ch[p][c - 'a'];
29             if (len[p] + 1 == len[q]) { // q is suffix of cur
30                 link[cur] = q;
31             } else { // we need auxiliary state
32                 int nxt = newnode();
33                 len[nxt] = len[p] + 1, link[nxt] = link[q];
34                 pos[nxt] = 0;
35                 for (int j = 0; j < C; j++)
36                     ch[nxt][j] = ch[q][j];
37                 while (~p && ch[p][c - 'a'] == q)
38                     ch[p][c - 'a'] = nxt, p = link[p];
39                 link[q] = link[cur] = nxt;
40             }
41             cnt[cur]++, last = cur;
42         }
43     } sam;
44 // void build_cnt() {
45 //     vector<int> p(_id); iota(all(p), 0); sort(all(p),
46 //         [&](int i, int j){ return len[i] > len[j]; });
47 //     for (int i = 0; i < _id; ++i) if (link[p[i]] != -1)
48 //         cnt[link[p[i]]] += cnt[p[i]];
49 // }
50 // bool match(string s) {
51 //     int p = 0; for (auto& c : s) {
52 //         if (!ch[p][c - 'a']) return false;
53 //         p = ch[p][c - 'a'];
54 //     } return true; }

```

10.7 Minimum Rotation

ae667e

```

1 // Author: std_abs
2 string rotate(auto s) { int n=s.size(), i=0, j=1;
3     string t = s + s;
4     while (i < n && j < n) {
5         int k = 0; while (k < n && t[i + k] == t[j + k]) ++k;
6         if (t[i + k] <= t[j + k]) j += k + 1; else i += k + 1;
7         if (i == j) ++j;
8     } return t.substr((i < n ? i : j), n);
9 }

```

10.8 AC Automaton

dc83a2

```

1 // Author: std_abs (slightly modified by Gino)
2 // Recap: fail[s] = s 在 trie 上 (也就是某個 pattern 的前綴
3 // 的最長 proper 後綴
4 // N := sum(length of all patterns) + 1
5 // 1. ac.insert(pat) -> end-node id recorded in ac.pid
6 // 2. ac.build_fail()
7 // 3. run text: ptr = to[ptr][c-'a'] each char;
8 //     accumulate on your own array
9 // - pull: rep(i ac.order_cnt-1 ~ 0) val[fail[x]] += val[x]
10 // - push: rep(i 0 ~ ac.order_cnt-1) val[x] += val[fail[x]]
11 // Tested @yosupo, N <= 1e6, 323ms on building AC
12 const int N = 1e6 + 5, C = 26;
13 struct AC {
14     int ch[N][C], to[N][C], fail[N], _id, order[N], order_cnt;
15     int np; vector<int> pid; // pattern ID on trie
16     // order[r] := which trie node has BFS order r
17     // fail link tree: (parent) fail[i] <--- i (child)
18     void reset() { _id = np = order_cnt = 0, pid.clear(),
19         newnode(); }
19     int newnode() {
20         fill_n(ch[_id], C, 0), fill_n(to[_id], C, 0);
21         fail[_id] = 0; return _id++;
22     }
23     AC () { reset(); }
24     void insert(string s) {
25         int ptr = 0; for (char c : s) {
26             if (!ch[ptr][c - 'a']) ch[ptr][c - 'a'] = newnode();
27             ptr = ch[ptr][c - 'a'];
28         } np++; pid.emplace_back(ptr);
29     }
30     void build_fail() {
31         queue<int> q; order_cnt = 0;
32         for (int i = 0; i < C; ++i) if (ch[0][i])

```

```

33         q.push(ch[0][i]), to[0][i] = ch[0][i];
34         while (!q.empty()) {
35             int u = q.front(); q.pop(); order[order_cnt++] = u;
36             for (int i = 0; i < C; ++i) {
37                 if (!ch[u][i]) to[u][i] = to[fail[u]][i];
38                 else { int v = ch[u][i], k = fail[u];
39                     while (k && !ch[k][i]) k = fail[k];
40                     if (ch[k][i]) k = ch[k][i];
41                     fail[v] = k, to[u][i] = v;
42                     q.push(v);
43                 }
44             }
45         } ac;

```

10.9 AC Automaton - Dict Links

009582

```

1 // Author: Gino, paste this in `struct AC {...}`
2 // 3 Modifications to the original AC Automaton:
3 // int np; vector<int> pid; bool is_pattern[N];
4 // fail[_id] = 0, is_pattern[_id] = false; return _id++;
5 // np++; pid.emplace_back(ptr); is_pattern[ptr] = true;
6 int dict[N];
7 void build_dict() { // call this after build_fail()!
8     for (int i = 0; i < order_cnt; i++) {
9         int u = order[i], f = fail[u];
10         dict[u] = is_pattern[f] ? f : dict[f];
11     }

```

10.10 AC Automaton - Counting Patterns

05fa55

```

1 // Author: Gino, paste this in `struct AC {...}`
2 ll val[N];
3 void count_patterns(string& text, vector<string& patterns){
4     for (auto& p : patterns) insert(p); build_fail();
5     ranges::fill(val, 0LL);
6     int ptr = 0; for (auto& c : text) {
7         ptr = to[ptr][c - 'a']; val[ptr]++;
8     }
9     for (int x : span(order, order_cnt) | views::reverse)
10         val[fail[x]] += val[x];
11     for (auto& i : pid) cout << val[i] << endl;
12 }

```

10.11 AC Automaton - Pattern Position

輸出每個 pattern 在 text 中第一次出現的位置，若不存在則輸出 -1。
走 AC 自動機時 if (val[ptr] == -1) val[ptr] = i; 接著 pull DP
if (val[x] == -1) continue;
if (val[fail[x]] == -1) val[fail[x]] = val[x];
else val[fail[x]] = min(val[fail[x]], val[x]);

11 Geometry

11.1 Template

de6583

```

1 using T = long long; // or double
2 const double eps = 1e-8;
3 int sgn(T x) { return (x > eps) - (x < -eps); }
4 struct Pt {
5     T x = 0, y = 0;
6     Pt operator+(Pt a) const { return {x + a.x, y + a.y}; }
7     Pt operator-(Pt a) const { return {x - a.x, y - a.y}; }
8     Pt operator*(T k) const { return {x * k, y * k}; }
9     Pt operator/(T k) const { return {x / k, y / k}; }
10    T operator*(Pt a) const { return x * a.x + y * a.y; }
11    T operator^(Pt a) const { return x * a.y - y * a.x; }
12    bool operator<(Pt a) const
13    { return tie(x, y) < tie(a.x, a.y); }
14    bool operator==(Pt a) const
15    { return !sgn(x - a.x) && !sgn(y - a.y); }
16    //friend ostream& operator<<(ostream& o, Pt a)
17    //{ return o << "(" << a.x << ", " << a.y << ")"; }
18};
19 struct Line { Pt a, b; }; // a -> b
20 T abs2(Pt a) { return a * a; }
21 double abs(Pt a) { return sqrt(abs2(a)); }
22 int ori(Pt o, Pt a, Pt b) // 1: CCW, -1: CW, 0: collinear
23 { return sgn((a - o) ^ (b - o)); }

```

11.2 Basic

b94994

```

1 struct Cir { Pt o; double r; };
2 // ----- 極角排序 -----
3 int ud(Pt a) // 0: upper half (incl. +x axis), 1: lower
4 { return !sgn(a.y) ? sgn(a.x) < 0 : a.y < 0; }
5 bool cmp(Pt a, Pt b) // polar angle in [0, 2PI),
6 { return ud(a) == ud(b) ? sgn(a ^ b) > 0 : ud(a) < ud(b); }
7 ranges::sort(p, [&](Pt a, Pt b){ return cmp(a - o, b - o); });
8 // -----
9 bool sameVec(Pt a, Pt b, int d)//d=0: parallel, d=1: same dir
10 { return !sgn(a ^ b) && sgn(a * b) > d * 2 - 2; }
11 bool sameVec(Line a, Line b, int d)
12 { return sameVec(a.b - a.a, b.b - b.a, d); }
13 Pt rot90(Pt a) { return {-a.y, a.x}; } // CCW
14 Pt ref(Pt a) { return ud(a) ? Pt(-a.x, -a.y) : a; }
15 T dbarea(vector<Pt>& p) { // doubled signed area, CCW > 0
16     T res = 0; for (int i = 0; i < sz(p); i++)

```

```

17   res += p[i] ^ p[(i+1) % sz(p)];
18   return res; }
19 bool onseg(Pt p, Pt a, Pt b) // p on segment ab?
20 { return !ori(p, a, b) && sgn((a - p) * (b - p)) <= 0; }
21 // ---- T = double ----
22 const double PI = acos(-1);
23 double normalize(double x) { // to [0, 2pi)
24   while (x < -eps) x += PI * 2;
25   while (x > PI * 2 + eps) x -= PI * 2;
26   return x; }
27 double theta(Pt a) { return normalize(atan2(a.y, a.x)); }
28 Pt unit(Pt a) { return a / abs(a); }
29 Pt rot(Pt a, double o) { // CCW
30   double c = cos(o), s = sin(o);
31   return Pt(c * a.x - s * a.y, s * a.x + c * a.y); }
32 Pt projVec(Pt a, Pt b, Pt c) // vector ac proj to ab
33 { return (b - a) * ((c - a) * (b - a)) / abs2(b - a); }
34 Pt projPt(Pt a, Pt b, Pt c) // point c proj to ab
35 { return projVec(a, b, c) + a; }

```

11.3 Line / Segment Intersections

471854

```

1 // m=0: segment, m=1: ray from l.a to l.b, m=2: line
2 bool banana(Line ls, int ms, Line lt, int mt, int strict) {
3   auto on = [&](Line l, int m, Pt p) {
4     if (ori(l.a, l.b, p) != 0) return false;
5     if (m && abs2(l.a - p) > abs2(l.b - p)) return true;
6     return m == 2 || sgn((p - l.a) * (p - l.b)) <= -strict;
7   };
8   if (sameVec(ls, lt, 0)) {
9     return on(ls, ms, lt.a) || on(ls, ms, lt.b) ||
10        on(lt, mt, ls.a) || on(lt, mt, ls.b);
11   }
12   auto good = [&](Line l, int m, Line o) {
13     if (m && abs((l.a - o.a) ^ (l.a - o.b)) >
14         abs((l.b - o.a) ^ (l.b - o.b))) return true;
15     return m == 2 ||
16            ori(l.a, o.a, o.b) * ori(l.b, o.a, o.b) == -1;
17   };
18   if (good(ls, ms, lt) && good(lt, mt, ls)) return 1;
19   if (!strict) {
20     if (mt != 2 && on(ls, ms, lt.a)) return 1;
21     if (mt == 0 && on(ls, ms, lt.b)) return 1;
22     if (ms != 2 && on(lt, mt, ls.a)) return 1;
23     if (ms == 0 && on(lt, mt, ls.b)) return 1;
24   }
25   return 0;
26 }
27 // becareful when two lines are parallel
28 auto bananaPoint(Line a, Line b) {
29   auto abc = (a.b - a.a) ^ (b.a - a.a);
30   auto abd = (a.b - a.a) ^ (b.b - a.a);
31   return make_pair(b.b * abc - b.a * abd, abc - abd);
32 }

```

11.4 Point-Line / Segment Distance

6b5619

```

1 // T = double
2 double pointLineDist(Pt p, Line L)
3 { return fabs(((L.b - L.a) ^ (p - L.a)) / abs(L.b - L.a)); }
4 double pointSegDist(Pt p, Pt a, Pt b) {
5   if (a == b) return abs(p - a);
6   if (sgn((b - a) * (p - a)) >= 0 &&
7       sgn((a - b) * (p - b)) >= 0)
8     return pointLineDist(p, {a, b});
9   return min(abs(p - a), abs(p - b));
10 }

```

11.5 Sort

59d0a3

```

1 // Dependency: cmp(), btwAngle()
2 // cmp in Basic: polar angle sort
3 // all points are on line ab. closer to a: front
4 bool cmpLine(Pt s, Pt t, Pt a, Pt b) {
5   Pt v = a - b;
6   if (sgn(v.x)) return sgn(s.x - t.x) == sgn(v.x);
7   return sgn(s.y - t.y) == sgn(v.y);
8 }
9 // intersect points polar angle sort, deno: positive
10 bool cmpFractionPolar(pair<Pt, ll> o,
11   pair<Pt, ll> s, pair<Pt, ll> t) {
12   Pt u = s.first * o.second - o.first * s.second;
13   Pt v = t.first * o.second - o.first * t.second;
14   return cmp(u, v);
15 }
16 struct Seg {
17   Pt a, b; // a.x < b.x
18   bool operator<(const Seg& o) const {
19     if (a == o.a) return ori(o.b, a, b) == 1;
20     if (a.x <= o.a.x) return ori(o.a, a, b) == 1;
21     return ori(a, o.a, o.b) == -1;
22   };
23   struct PolarSeg {

```

```

24   Pt a, b; // ori(Pt(0, 0), a, b) > 0
25   bool operator<(const PolarSeg& o) const {
26     if (a == o.a) return ori(o.b, a, b) == -1;
27     if (btwAngle(Pt(0, 0), a, b, o.a, 0))
28       return ori(o.a, a, b) == -1;
29     return ori(a, o.a, o.b) == 1;
30   };
31   struct Arc {
32     // contain(a, b): circle a inside circle b
33     Cir c; int id, s; // 0 -> up, 1 -> down
34     bool operator<(const Arc& b) const {
35       if (id == b.id) return s < b.s;
36       if (contain(c, b.c)) return b.s == 1;
37       if (contain(b.c, c)) return s == 0;
38       if (c.o.y == b.c.o.y) return id < b.id;
39       return c.o.y > b.c.o.y;
40     };

```

11.6 Convex Hull

788ba8

```

1 auto convexHull(vector<Pt> pts) {
2   sort(all(pts)); vector<Pt> ans = {pts[0]};
3   for (int t = 0; t < 2; t++, reverse(all(pts))) {
4     for (int i = 1, m = sz(ans); i < sz(pts); i++) {
5       while (sz(ans) > m &&
6             ori(ans[sz(ans) - 2], ans.back(), pts[i]) <= 0)
7         ans.pop_back();
8       ans.pb(pts[i]); } }
9   if (sz(ans) > 1) ans.pop_back(); return ans; }

```

11.7 Triangle Centers

12240b

```

1 // T = double
2 Pt circenter(Pt a, Pt b, Pt c) { // r = abs(o - a)
3   b = b - a, c = c - a; double m = 2 * (b^c);
4   Pt o = {(abs2(b) * c.y - abs2(c) * b.y) / m,
5           (abs2(c) * b.x - abs2(b) * c.x) / m};
6   return o + a;
7 }
8 Pt incenter(Pt a, Pt b, Pt c) { // r = 2 * area / s
9   double x = abs(b - c), y = abs(c - a);
10  double z = abs(a - b), s = x + y + z;
11  return (a*x + b*y + c*z) / s;
12 }
13 Pt centroid(Pt a, Pt b, Pt c) { return (a + b + c) / 3; }
14 Pt orthocenter(Pt a, Pt b, Pt c) {
15   return centroid(a, b, c) * 3 - circenter(a, b, c) * 2;
16 }

```

11.8 Point Inside

670564

```

1 // get edge index: check (0, a), (0, b) first
2 // then after binary search, check (a, b)
3 bool inConvex(vector<Pt>& C, Pt p, bool strict = true) {
4   // only works when no three points are collinear
5   int a = 1, b = sz(C) - 1, r = !strict;
6   if (sz(C) == 0) return false;
7   if (sz(C) < 3) return r && onseg(p, C[0], C.back());
8   if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
9   if (ori(C[0], C[a], p) >= r ||
10       ori(C[0], C[b], p) <= -r) return false;
11   while (abs(a - b) > 1) {
12     int c = (a + b) / 2;
13     (ori(C[0], C[c], p) > 0 ? b : a) = c;
14   }
15   return ori(C[a], C[b], p) < r;
16 }
17 // need onseg(), -1: out, 0: edge, 1: in
18 int inPoly(vector<Pt> poly, Pt o, int strict) {
19   int cnt = 0;
20   for (int i = 0; i < sz(poly); i++) {
21     Pt a = poly[i], b = poly[(i + 1) % sz(poly)];
22     if (onseg(o, a, b)) return !strict;
23     cnt ^= ((o.y < a.y) - (o.y < b.y)) * ori(o, a, b) > 0;
24   }
25   return cnt ? 1 : -1;
26 }
27 // relation of q with circumcircle; T = long long
28 bool inCC(array<Pt, 3> p, Pt q) {
29   __int128 det = 0;
30   for (int i = 0; i < 3; i++)
31     det += __int128(abs2(p[i]) - abs2(q)) *
32           ((p[(i + 1) % 3] - q) ^ (p[(i + 2) % 3] - q));
33   return det > 0; // in: >0, on: =0, out: <0
34 }

```

11.9 Point In Triangle

0eabd1

```

1 // number of p with a < p < b and ori(p, a, b) < 0
2 int under(Pt a, Pt b) {}
3 // number of points with a < p < b and ori(p, a, b) = 0
4 int edge(Pt a, Pt b) {}
5 // check if this number is calculated
6 bool check(Pt p) {}
7 // number of points that strictly inside the triangle

```

```

8 int inTri(array<Pt, 3> arr) {
9     sort(all(arr)); auto [a, b, c] = arr; intx = ori(b, a, c);
10    if (x == 0) return 0;
11    if (x == 1) return under(a, b) + under(b, c) -
12        under(a, c) - edge(a, c);
13    return under(a, c) - under(a, b) -
14        under(b, c) - edge(a, b) - edge(b, c) - check(b);
15}

```

11.10 Vector In Polygon

970582

```

1 // ori(a, b, c) >= 0, valid: "strict" angle from a-b to a-c
2 bool btwAngle(Pt a, Pt b, Pt c, Pt p, int strict) {
3     return ori(a, b, p) >= strict &&
4         ori(a, p, c) >= strict;
5 }
6 // whether vector{cur, p} is in order prv, cur, nxt
7 bool inside(Pt prv, Pt cur, Pt nxt, Pt p, int strict) {
8     if (ori(cur, nxt, prv) >= 0)
9         return btwAngle(cur, nxt, prv, p, strict);
10    return !btwAngle(cur, prv, nxt, p, !strict);
11}

```

11.11 Polygon Cut

21751d

```

1 // Dependency: bananaPoint(); T = double
2 // Cut a convex polygon with line (s -> e), keep right side
3 vector<Pt> polyCut(vector<Pt> poly, Pt s, Pt e) {
4     vector<Pt> res; for (int i = 0; i < sz(poly); i++) {
5         Pt cur = poly[i], prv = i ? poly[i - 1] : poly.back();
6         bool side = ori(s, e, cur) < 0;
7         if (side != (ori(s, e, prv) < 0)) {
8             auto [p, d] = bananaPoint({s, e}, {cur, prv});
9             res.pb(p / d);
10        } if (side) res.pb(cur);
11    } return res;
12}

```

11.12 Half-Plane Intersection

1cf136

```

1 // Dependency: sameVec(), cmp(); T = long long
2 /* Having solution, check size > 2 */
3 /* -- Line.a -- Line.b -- */
4 auto halfPlaneIntersection(vector<Line> arr) {
5     auto areaPair = [&](Line a, Line b) {
6         return make_pair((a.b - a.a) ^ (b.a - a.a),
7             (a.b - a.a) ^ (b.b - a.a));
8     };
9     auto isin = [&](Line l0, Line l1, Line l2) {
10        // Check inter(l1, l2) strictly in l0
11        auto [a02X, a02Y] = areaPair(l0, l2);
12        auto [a12X, a12Y] = areaPair(l1, l2);
13        if (a12X - a12Y < 0) a12X *= -1, a12Y *= -1;
14        return (__int128)a02Y * a12X -
15            (__int128)a02X * a12Y > 0; // C^4
16    };
17    sort(all(arr), [&](Line a, Line b) {
18        if (sameVec(a, b, 1))
19            return ori(a.a, a.b, b.b) < 0;
20        return cmp(a.b - a.a, b.b - b.a);
21    });
22    deque<Line> dq(1, arr[0]);
23    auto popBack = [&](int t, Line p) {
24        while (sz(dq) >= t &&
25            !isin(p, dq[sz(dq) - 2], dq.back()))
26            dq.pop_back();
27    };
28    auto popFront = [&](int t, Line p) {
29        while (sz(dq) >= t && !isin(p, dq[0], dq[1]))
30            dq.pop_front();
31    };
32    for (auto p : arr) if (!sameVec(dq.back(), p, 1))
33        popBack(2, p), popFront(2, p), dq.pb(p);
34    popBack(3, dq[0]), popFront(3, dq.back());
35    return vector<Line>(all(dq));
36}

```

11.13 Cyclic Ternary Search

d1b304

```

1 /* bool pred(int a, int b);
2 f(0) ~ f(n - 1) is a cyclic-shift U-function
3 return idx s.t. pred(x, idx) is false forall x*/
4 int cycTsearch(int n, auto pred) {
5     if (n == 1) return 0;
6     int l = 0, r = n; bool rv = pred(1, 0);
7     while (r - l > 1) {
8         int m = (l + r) / 2;
9         if (pred(0, m) ? rv : pred(m, (m + 1) % n)) r = m;
10        else l = m;
11    } return pred(l, r % n) ? l : r % n;
12}

```

11.14 Circle Intersections

8f97aa

```

1 // Dependency: cycTsearch(); T = double for circles
2 vector<Pt> circleLineIntersect(Cir c, Line l) {

```

```

3     Pt p = l.a + (l.b - l.a) *
4         ((c.o - l.a) * (l.b - l.a)) / abs2(l.b - l.a);
5     double s = (l.b - l.a) ^ (c.o - l.a);
6     double h2 = c.r * c.r - s * s / abs2(l.b - l.a);
7     if (sgn(h2) == -1) return {};
8     if (sgn(h2) == 0) return {p};
9     Pt h = (l.b - l.a) / abs(l.b - l.a) * sqrt(h2);
10    return {p - h, p + h};
11}
12 // covered area of c1: arc from res[0] to res[1], CCW
13 vector<Pt> circlesIntersect(Cir c1, Cir c2) {
14    double d2 = abs2(c1.o - c2.o), d = sqrt(d2);
15    if (d < max(c1.r, c2.r) - min(c1.r, c2.r) ||
16        d > c1.r + c2.r) return {};
17    Pt u = (c1.o + c2.o) / 2 +
18        (c1.o - c2.o) *
19        ((c2.r * c2.r - c1.r * c1.r) / (2 * d2));
20    double A = sqrt((c1.r + c2.r + d) *
21        (c1.r - c2.r + d) * (c1.r + c2.r - d) *
22        (-c1.r + c2.r + d));
23    Pt v = rot90(c2.o - c1.o) * A / (2 * d2);
24    if (sgn(v.x) == 0 && sgn(v.y) == 0) return {u};
25    return {u - v, u + v};
26}
27 vector<pii> convexLineIntersect(vector<Pt> C, Line la) {
28    auto dis = [&](int p) {
29        return (la.b - la.a) ^ (C[p] - la.a);
30    };
31    auto gao = [&](int s) {
32        return cycTsearch(sz(C), [&](int i, int j) {
33            return sgn(dis(i) - dis(j)) == s;
34        });
35    };
36    int x = gao(1), y = gao(-1), n = sz(C);
37    if (sgn(dis(x)) < 0 || sgn(dis(y)) > 0) return {};
38    if (sgn(dis(x)) == 0 || sgn(dis(y)) == 0) {
39        int v = ((sgn(dis(x)) == 0 ? x : y) + n - 1) % n;
40        vector<pii> vec;
41        for (int i = 0; i < 3; i++, v = (v + 1) % n)
42            if (sgn(dis(v)) == 0) vec.emplace_back(v, v);
43        return vec;
44    }
45    auto get = [&](int l, int r, int s) {
46        while ((l + 1) % n != r) {
47            int m = ((l + r + (l < r ? 0 : n)) / 2) % n;
48            if (sgn(dis(m)) == s ? l : r) = m;
49        }
50        if (sgn(dis(r)) == 0) return pii(r, r);
51        return pii(l, r);
52    };
53    return {get(x, y, 1), get(y, x, -1)};
54}

```

11.15 Tangents

7d0503

```

1 // Dependency: cycTsearch(); T = double for circle tangents
2 auto circlePointTangent(Cir c, Pt p) {
3     vector<Line> res;
4     double dSq = abs2(p - c.o);
5     if (sgn(dSq - c.r * c.r) == 0) {
6         res.push_back({p, p + rot90(p - c.o)});
7     } else if (dSq > c.r * c.r) {
8         double s = dSq - c.r * c.r;
9         Pt v = p + (c.o - p) * s / dSq;
10        Pt u = rot90(c.o - p) * sqrt(s) * c.r / dSq;
11        res.push_back({p, v + u});
12        res.push_back({p, v - u});
13    }
14    return res;
15}
16 auto circlesTangent(Cir c1, Cir c2, int sign1) {
17    // sign1 = 1 for outer tang, -1 for inner tang
18    vector<Line> res;
19    double dSq = abs2(c1.o - c2.o);
20    if (sgn(dSq) == 0) return res;
21    double d = sqrt(dSq);
22    Pt v = (c2.o - c1.o) / d;
23    double c = (c1.r - sign1 * c2.r) / d;
24    if (c * c > 1) return res;
25    double h = sqrt(max(0.0, 1.0 - c * c));
26    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
27        Pt n = Pt(v.x * c - sign2 * h * v.y,
28            v.y * c + sign2 * h * v.x);
29        Pt p1 = c1.o + n * c1.r;
30        Pt p2 = c2.o + n * (c2.r * sign1);
31        if (sgn(p1.x - p2.x) == 0 &&
32            sgn(p1.y - p2.y) == 0)
33            p2 = p1 + rot90(c2.o - c1.o);
34        res.push_back({p1, p2});
35    }

```

```

36 return res;
37 }
38 /* The point should be strictly out of hull
39 return arbitrary point on the tangent line */
40 pii pointConvexTangent(vector<Pt>& C, Pt p) {
41     auto gao = [&](int s) {
42         return cycTsearch(sz(C), [&](int x, int y) {
43             return ori(p, C[x], C[y]) == s;
44         });
45     };
46     return pii(gao(1), gao(-1));
47 } // return (a, b), ori(p, C[a], C[b]) >= 0

```

11.16 Bisector

a45c44

```

1 Pt bisector(Pt a, Pt b, Pt c) { Pt s = b-a, t = c-a;
2   t = t * abs(s) / abs(t); return s + t; }

```

11.17 Minimum Enclosing Circle

cd2fc9

```

1 // Dependency: circenter(); T = double
2 Cir minEnclosingCircle(vector<Pt> p) {
3     auto seed = chrono::steady_clock::now();
4     time_since_epoch().count();
5     shuffle(all(p), mt19937(seed));
6     double r = 0.0; Pt cent = p[0];
7     for (int i = 1; i < sz(p); i++) {
8         if (abs2(cent - p[i]) <= r) continue;
9         cent = p[i]; r = 0.0;
10        for (int j = 0; j < i; j++) {
11            if (abs2(cent - p[j]) <= r) continue;
12            cent = (p[i] + p[j]) / 2; r = abs2(p[j] - cent);
13            for (int k = 0; k < j; k++) {
14                if (abs2(cent - p[k]) <= r) continue;
15                cent = circenter(p[i], p[j], p[k]);
16                r = abs2(p[k] - cent);
17            }
18        }
19    }
20    return {cent, sqrt(r)};

```

11.18 Minkowski Sum

133299

```

1 // Dependency: convexHull()
2 auto minkowski(vector<Pt> a, vector<Pt> b) {
3     a = convexHull(a), b = convexHull(b);
4     int n = sz(a), m = sz(b);
5     vector<Pt> c = {a[0] + b[0]}, s1, s2;
6     for (int i = 0; i < n; i++) s1.pb(a[(i+1) % n] - a[i]);
7     for (int i = 0; i < m; i++) s2.pb(b[(i+1) % m] - b[i]);
8     for (int p1 = 0, p2 = 0; p1 < n || p2 < m;)
9         if (p2 == m || (p1 < n && sgn(s1[p1] ^ s2[p2]) >= 0))
10            c.pb(c.back() + s1[p1++]);
11        else
12            c.pb(c.back() + s2[p2++]);
13    return convexHull(c);

```

11.19 Convex Hull Distance

987fcf

```

1 // Dependency: minkowski(), inConvex(), pointSegDist()
2 // T = double
3 double convexHullDist(vector<Pt> A, vector<Pt> B) {
4     Pt O(0, 0); for (auto& p : B) p = O - p;
5     auto C = minkowski(A, B);
6     if (inConvex(C, O)) return 0;
7     double ans = pointSegDist(O, C.back(), C[0]);
8     for (int i = 0; i + 1 < sz(C); i++)
9         ans = min(ans, pointSegDist(O, C[i], C[i + 1]));
10    return ans; }

```

11.20 Closest Point Pair

45eca8

```

1 // returns {min squared dist, {i, j}}, n >= 2, |coord| < 1e9
2 pair<T, pair<int, int>> closestPair(vector<Pt> p) {
3     vector<int> id(sz(p)); iota(all(id), 0);
4     sort(all(id),
5         [&](int a, int b) { return p[a].y < p[b].y; });
6     set<pair<Pt, int>> s; // (point, id)
7     pair<T, pair<int, int>> best = {(T)9e18, {-1, -1}};
8     int l = 0; for (int i : id) {
9         if (!best.first) break;
10        T d = sqrtl(best.first) + is_integral_v<T>;
11        while (p[id[l]].y <= p[i].y - d)
12            s.erase({p[id[l]], id[l]}), l++;
13        auto lo = s.lower_bound({p[i] - Pt{d, 0}, 0});
14        auto hi = s.upper_bound({p[i] + Pt{d, 0}, sz(p)});
15        for (; lo != hi; lo++) {
16            auto [q, j] = *lo;
17            best = min(best,
18                {abs2(q - p[i]), {min(i, j), max(i, j)}});
19        } s.insert({p[i], i});
20    } return best;

```

11.21 Polygon-Circle Intersection

51ecf1

```

1 // T = double

```

```

2 double areaSector(Pt pa, Pt pb, double r) {
3     if (abs(pa) < abs(pb)) swap(pa, pb);
4     if (abs(pb) < eps) return 0;
5     double S, h, t;
6     double a = abs(pb), b = abs(pa), c = abs(pb - pa);
7     double cosB = pb * (pb - pa) / a / c;
8     double B = acos(cosB);
9     double cosC = pa * pb / a / b, C = acos(cosC);
10    if (a > r) {
11        S = C / 2 * r * r;
12        h = a * b * sin(C) / c;
13        if (h < r && B < PI / 2)
14            S -= acos(h / r) * r * r - h * sqrt(r * r - h * h);
15    } else if (b > r) {
16        t = PI - B - asin(sin(B) / r * a);
17        S = 0.5 * a * r * sin(t) + (C - t) / 2 * r * r;
18    } else {
19        S = 0.5 * sin(C) * a * b;
20    }
21    return S;
22 }
23 double areaPolyCircle(vector<Pt> poly, Pt O, double r) {
24     double S = 0;
25     int n = sz(poly);
26     for (int i = 0; i < n; i++)
27         S += areaSector(poly[i] - O,
28             poly[(i + 1) % n] - O, r) *
29             ori(O, poly[i], poly[(i + 1) % n]);
30    return fabs(S);
31 }

```

11.22 Union of Polygons

f5b117

```

1 // Author: Gino, simple polygons any orientation, O((sum n)^2)
2 // T = ll: exact (|coord| <= 1e9); T = double: also works
3 double polysUnionArea(vector<vector<Pt>> poly) {
4     struct Fr { T p, q; }; // p / q, q > 0
5     auto lt = [](Fr a, Fr b) {
6         return ( (__int128)a.p * b.q < ( __int128)b.p * a.q; // if T
7             == long long
8             return a.p * b.q < b.p * a.q;
9         };
10    auto val = [&](Fr a) { // clamp to [0, 1], to double
11        if (lt(a, {0, 1})) return 0.;
12        if (lt({1, 1}, a)) return 1.;
13        return (double)a.p / a.q;
14    };
15    auto get = [&](int i, int e) {
16        return poly[i][(e + 1) % sz(poly[i])];
17    };
18    for (auto& p : poly)
19        if (dbarea(p) < 0) reverse(all(p));
20    double sum = 0;
21    for (int i = 0; i < sz(poly); i++)
22        for (int e = 0; e < sz(poly[i]); e++) {
23            Pt s = poly[i][e], t = get(i, e);
24            vector<pair<Fr, int>> c = {{0, 1}, 0}, {{1, 1}, 0}};
25            for (int j = 0; j < sz(poly); j++) {
26                if (i == j) continue;
27                for (int f = 0; f < sz(poly[j]); f++) {
28                    Pt a = poly[j][f], b = get(j, f);
29                    int ta = ori(s, t, a), tb = ori(s, t, b);
30                    if (!ta && !tb) {
31                        if (j < i && sgn((b-a) * (t-s)) > 0) {
32                            T d = abs2(t-s);
33                            c.pb({Fr{(a-s) * (t-s), d}, 1});
34                            c.pb({Fr{(b-s) * (t-s), d}, -1});
35                        }
36                    } else if ((ta >= 0) != (tb >= 0)) {
37                        T x = (b-a) ^ (s-a), y = (b-a) ^ (t-a);
38                        if (x < y) x = -x, y = -y;
39                        c.pb({Fr{x, x-y}, ta >= 0 ? 1 : -1});
40                    }
41                }
42            }
43            sort(all(c), [&](auto& u, auto& v) {
44                return lt(u.first, v.first); });
45            double z = val(c[0].first), r = 0;
46            for (int k = 1, d = c[0].second; k < sz(c); k++) {
47                double w = val(c[k].first);
48                if (!d) r += w - z;
49                d += c[k].second, z = w;
50            }
51            sum += (s^t) * r;
52        }
53    return sum / 2;

```

11.23 Rotating Sweep Line

3b614c

```

1 // Dependency: ref()

```



```

2 struct Event {
3     Pt d; int u, v;
4     bool operator<(const Event& o) const {
5         return sgn(d ^ o.d) > 0;
6     };
7 void rotatingSweepLine(vector<Pt> pts) {
8     int n = sz(pts);
9     vector<int> ord(n), pos(n); vector<Event> e;
10    for (int i = 0; i < n; i++)
11        for (int j = i + 1; j < n; j++)
12            e.push_back({ref(pts[i] - pts[j]), i, j});
13    sort(all(e)); iota(all(ord), 0);
14    sort(all(ord), [&](int i, int j) {
15        return sgn(pts[i].y - pts[j].y) == 0 ?
16            pts[i].x < pts[j].x : pts[i].y < pts[j].y;
17    });
18    for (int i = 0; i < n; i++) pos[ord[i]] = i;
19    auto makeReverse = [](auto v) {
20        sort(all(v));
21        v.resize(unique(all(v)) - v.begin());
22        vector<pii> segs;
23        for (int i = 0, j = 0; i < sz(v); i = j) {
24            for (; j < sz(v) && v[j].y - v[i].y <= j - i; j++);
25            segs.emplace_back(v[i], v[j - 1] + 1 + 1);
26        }
27        return segs;
28    };
29    for (int i = 0, j = 0; i < sz(e); i = j) {
30        vector<int> tmp;
31        for (; j < sz(e) && !(e[i] < e[j]); j++);
32        tmp.pb(min(pos[e[j].u], pos[e[j].v]));
33        for (auto [l, r] : makeReverse(tmp)) {
34            reverse(ord.begin() + l, ord.begin() + r);
35            for (int t = l; t < r; t++) pos[ord[t]] = t;
36            // update value here
37        }
38    }

```

11.24 Pick's Theorem

Consider a polygon which vertices are all lattice points.
 Let i = number of points inside the polygon.
 Let b = number of points on the boundary of the polygon.
 Then we have the following formula:

$$\text{Area} = i + b/2 - 1$$

12 Math Basic

12.1 Basic

8881cf

```

1 const ll mod = 998244353;
2 ll add(ll a, ll b) { return (a += b) >= mod ? a - mod : a; }
3 ll sub(ll a, ll b) { return (a -= b) < 0 ? a + mod : a; }
4 ll mul(ll a, ll b) { return (__int128)a * b % mod; }
5 ll Pow(ll b, ll e) { ll r = 1;
6     for (; e >= 1; b = mul(b, b)) if (e & 1) r = mul(r, b);
7     return r;
8 }

```

13 Number Theory

13.1 Mod Sum / Floor Sum

c72030

```

1 // modSum(to, c, k, m) sums (k*i+c) mod m for i < to.
2 // divSum computes the corresponding sum of floor((k*i+c)/m).
3 // Both run in O(log m).
4 using ull = unsigned long long;
5
6 ull sumsq(ull to) { return to / 2 * ((to - 1) | 1); }
7
8 ull divSum(ull to, ull c, ull k, ull m) {
9     ull res = k / m * sumsq(to) + c / m * to;
10    k %= m;
11    c %= m;
12    if (!k) return res;
13    ull to2 = (to * k + c) / m;
14    return res + (to - 1) * to2
15        - divSum(to2, m - 1 - c, m, k);
16 }
17
18 ll modSum(ull to, ll c, ll k, ll m) {
19     c = ((c % m) + m) % m;
20     k = ((k % m) + m) % m;
21     return to * c + k * sumsq(to) - m * divSum(to, c, k, m);
22 }

```

13.2 Mod Sqrt

79b959

```

1 // sqrtMod(a, p) finds x with x^2 = a (mod p), p prime
2 // 2 Roots: (x, p - x), returns the smaller one
3 // (Tonelli-Shanks) O(log^2 p), usually O(log p).
4 // Pow(b, e, p): b^e mod p
5 ll sqrtMod(ll a, ll p) { a = (a % p + p) % p;
6     if (p == 2) return a;

```

```

7     if (a == 0) return 0;
8     if (Pow(a, (p - 1) / 2, p) != 1) return -1;
9     if (p % 4 == 3) {
10        ll x = Pow(a, (p + 1) / 4, p); return min(x, p - x); }
11    ll s = p - 1, n = 2;
12    int r = 0, m;
13    while (s % 2 == 0) r++, s /= 2;
14    while (Pow(n, (p - 1) / 2, p) != p - 1) n++;
15    ll x = Pow(a, (s + 1) / 2, p);
16    ll b = Pow(a, s, p), g = Pow(n, s, p);
17    for (; r = m) {
18        ll t = b;
19        for (m = 0; m < r && t != 1; m++) t = t * t % p;
20        if (m == 0) return min(x, p - x);
21        ll gs = Pow(g, 1LL << (r - m - 1), p);
22        g = gs * gs % p;
23        x = x * gs % p;
24        b = b * g % p;
25    }

```

13.3 Discrete Log (exBSGS)

f4ee37

```

1 // Find the smallest N s.t. X^N = Y (mod P)
2 #define rep(i, a, b) for (int i = a; i <= b; i++)
3 ll Log(ll X, ll Y, ll P) { // X, Y, P any
4     X %= P, Y %= P; ll D = 1; int add = 0;
5     while (true) {
6         const auto g = gcd(X, P);
7         if (g == 1) break;
8         if (D == Y) return add;
9         if (Y % g != 0) return -1;
10        Y /= g, P /= g, add++;
11        D = D * (X / g) % P;
12    }
13    const ll B = sqrtl(P) + 1;
14    unordered_map<ll, int> mp;
15    for (ll i = 0, cur = Y; i <= B; i++) {
16        mp[cur] = i;
17        cur = cur * X % P;
18    }
19    ll step = 1;
20    rep(i, 1, B) step = step * X % P;
21    for (ll p = 1, cur = D; p <= B; p++) {
22        cur = cur * step % P;
23        if (mp.contains(cur)) {
24            return B * p - mp[cur] + add;
25        }
26    }
27    return -1;

```

13.4 Miller Rabin

bd7351

```

1 // isPrime(n): deterministic for n < 2^63; 1e6 calls ~0.25s
2 // mul(a, b, p): (__int128)a * b % p;
3 // Pow(b, e, p): b^e mod p
4 bool isPrime(ll n) {
5     if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
6     ll A[] = {2, 325, 9375, 28178, 450775, 9780504,
7         1795265022};
8     ll s = __builtin_ctzll(n - 1), d = n >> s;
9     for (ll a : A) {
10        ll p = Pow(a % n, d, n), i = s;
11        while (p != 1 && p != n - 1 && a % n && i--)
12            p = mul(p, p, n);
13        if (p != n - 1 && i != s) return 0;
14    }
15    return 1;

```

13.5 Pollard Rho

ebf488

```

1 // factorization(x): sorted (p, e) with x = prod p^e, x < 2^62
2 // needs: MillerRabin. Avg time per x (-02, worst = semiprime):
3 // x~1e9: 0.003ms | x~1e12: 0.004ms, semi 0.012ms
4 // x~1e18: 0.016ms, semi 0.33ms (~3000 semiprimes / s)
5 ll pollard(ll n) {
6     ll x = 0, y = 0, t = 30, prd = 2, i = 1, q;
7     auto f = [&](ll x) { return mul(x, x, n) + i; };
8     while (t++ % 40 || gcd(prd, n) == 1) {
9         if (x == y) x = ++i, y = f(x);
10        if ((q = mul(prd, abs(x - y), n))) prd = q;
11        x = f(x), y = f(f(y));
12    }
13    return gcd(prd, n);
14 }
15 vector<pair<ll, int>> factorization(ll x) {
16     map<ll, int> cnt;
17     auto rec = [&](auto&& self, ll x) -> void {
18         if (x == 1) return;
19         if (isPrime(x)) { cnt[x]++; return; }
20         ll d = pollard(x);

```

```

21     self(self, d), self(self, x / d);
22 };
23 rec(rec, x);
24 return {all(cnt)};
25 }

```

13.6 Linear Sieve

$O(n)$, each i crossed out once by its least prime p :

```

for i in 2..n:
    if lpf[i] == 0: lpf[i] = i, primes.pb(i), f[i] = f(p)
    for p in primes:
        if p > lpf[i] or i * p > n: break
        lpf[i*p] = p
        f[i*p] = (i % p ? f[i] * f[p] : <recurrence>)

```

Recurrence for ip when $p \mid i$ (let $i = p^k m, p \nmid m$):

- $\varphi(ip) = \varphi(i) \cdot p$, $\mu(ip) = 0$, $\sigma_0(ip) = \sigma_0(i) \cdot \frac{k+2}{k+1}$
- General: keep $pw[i] = p^k$; $f(ip) = f\left(\frac{i}{pw[i]}\right) \cdot f(pw[i] \cdot p)$
- $\varphi(ip) = \varphi(i) \cdot (p - [p \nmid i])$ (one formula for both cases)

Euler phi table alone (no primes): $O(n \log \log n)$

for i: phi[i]=i; for p prime: for j=p,2p...: phi[j] -= phi[j]/p

13.7 Lucy (prime counting / prime sum)

1df76c

```

1 template<class T, class F>
2 pair<vector<T>, vector<T>> Lucy(const i64 N, const F f) {
3     const i64 sqrtN = sqrtl(N);
4     vector<T> Sl(sqrtN + 1);
5     vector<T> Sh(sqrtN + 1);
6     for (int i = 1; i <= sqrtN; i++) {
7         Sl[i] = f(i) - 1;
8         Sh[i] = f(f64(N) / i) - 1;
9     }
10    for (const int &p : primes) {
11        const i64 pp = i64(p) * p;
12        if (pp > N) break;
13        const int R = min(sqrtN, N / pp);
14        const int M = sqrtN / p;
15        const auto pi = Sl[p - 1];
16        const auto fp = Sl[p] - Sl[p - 1];
17        for (int i = 1; i <= M; i++) Sh[i] -= fp * (Sh[i * p] - pi);
18        for (int i = M + 1; i <= R; i++) Sh[i] -= fp * (Sl[N / (f64(i) * p)] - pi);
19        for (int i = sqrtN; i >= pp; i--) Sl[i] -= fp * (Sl[i / p] - pi);
20    }
21    return {Sl, Sh};
22 }

```

13.8 Black (Min25 sieve)

a652c9

```

1 template<class T, class F1, class F2>
2 T Black(const i64 N, const F1 f_prime, const F2 f_k) {
3     const i64 sqrtN = sqrtl(N);
4     const int np = int(size(primes));
5     T res = 1 + f_prime(N);
6     function<void(i64, i64, i64, T, T)> rec = [&](i64 t, i64 i, i64 c, T ft, T fu) {
7         const auto f_next = fu * f_k(primes[i], c + 1);
8         res += f_next + ft * (f_prime(f64(N) / t) - f_prime(primes[i]));
9         const i64 lim = sqrtl(f64(N) / t);
10        if (primes[i] <= lim) {
11            rec(t * primes[i], i, c + 1, f_next, fu);
12        }
13        for (int j = i + 1; j < np; j++) {
14            if (primes[j] > lim) break;
15            rec(t * primes[j], j, 1, ft * f_k(primes[j], 1), ft);
16        }
17    };
18    for (int i = 0; i < np; i++) {
19        if (primes[i] <= sqrtN) {
20            rec(primes[i], i, 1, f_k(primes[i], 1), 1);
21        }
22    }
23    return res;
24 }

```

13.9 Fast GCD

d55d7c

```

1 // O(1) gcd(x, y) for 1 <= x, y <= n after O(n) init.
2 // Use when ~1e7+ gcd queries on values <= n (e.g. sum of
3 // gcd(a[i], a[j]), gcd-convolution); else std::gcd is fine.
4 // Usage: GCD<N, SQRT> G; (global) G.init(n); G.fastGCD(x, y)
5 // N > n, SQRT > sqrt(n). Memory 20 bytes * N.
6 // -O2: n=1e6 init 17ms, 20MB; n=1e7 init 0.2s, 200MB.
7 // 1e7 random queries: 0.2s (std::gcd 0.6-0.7s, ~3x).
8 template <size_t N, size_t SQRT>
9 struct GCD {
10     int sq, minp[N], gcd_table[SQRT][SQRT];

```

```

11     vector<int> primes;
12     array<int, 3> fac[N];
13     void init(const int n) {
14         sq = int(sqrt(n));
15         for (int i = 0; i <= sq; ++i)
16             gcd_table[0][i] = gcd_table[i][0] = i;
17         for (int i = 1; i <= sq; ++i)
18             for (int j = 1; j <= i; ++j)
19                 gcd_table[i][j] = gcd_table[j][i] = gcd_table[j][i % i % j];
20         for (int i = 1; i <= n; ++i)
21             minp[i] = i;
22         fac[1] = {1, 1, 1};
23         for (int i = 2; i <= n; ++i) {
24             if (minp[i] == i) {
25                 primes.push_back(i);
26                 fac[i] = {1, 1, i};
27             }
28             for (const int &p : primes) {
29                 if (i * p > n) break;
30                 const int j = i * p;
31                 minp[j] = p;
32                 fac[j] = fac[i];
33                 fac[j][0] *= p;
34                 ranges::sort(fac[j]);
35                 if (i % p == 0) break;
36             }
37         }
38         int fastGCD(int x, int y) {
39             int g = 1;
40             for (const int &f : fac[x]) {
41                 int c = 1;
42                 if (f <= sq) c = gcd_table[f][y % f];
43                 else if (y % f == 0) c = f;
44                 y /= c; g *= c;
45             }
46             return g;
47         }
48     };

```

13.10 CRT

255819

```

1 // Solves x = a (mod m), x = b (mod n).
2 // m,n need not be coprime; asserts if no solution.
3 // Needs m*n < 2^62-ish; extgcd returns a*x+b*y=g.
4 ll extgcd(ll a, ll b, ll& x, ll& y) {
5     if (!b) return x = 1, y = 0, a;
6     ll g = extgcd(b, a % b, y, x);
7     return y -= a / b * x, g;
8 }
9
10 ll crt(ll a, ll m, ll b, ll n) {
11     if (n > m) swap(a, b), swap(m, n);
12     ll x, y, g = extgcd(m, n, x, y);
13     assert((a - b) % g == 0);
14     x = (b - a) % n * x % n / g * m + a;
15     return x < 0 ? x + m * n / g : x;
16 }

```

13.11 Extended Lucas

e73902

```

1 typedef long long ll;
2 void exgcd(ll a, ll b, ll& x, ll& y) {
3     if (!b)
4         x = 1, y = 0;
5     else
6         exgcd(b, a % b, y, x), y -= a / b * x;
7 }
8 ll inv(ll a, ll p) {
9     ll x, y;
10    exgcd(a, p, x, y);
11    return (x + p) % p;
12 }
13 ll qpow(ll a, ll b, ll p) {
14     ll r = 1;
15     for (; b >= 1; a = a * a % p)
16         if (b & 1) r = r * a % p;
17     return r;
18 }
19 ll CRT(int n, ll* a, ll* m) {
20     ll M = 1, p = 0;
21     for (int i = 1; i <= n; i++) M = M * m[i];
22     for (int i = 1; i <= n; i++) {
23         ll w = M / m[i], x, y;
24         exgcd(w, m[i], x, y);
25         p = (p + a[i] * w * x % M) % M;
26     }
27     return (p % M + M) % M;
28 }
29 ll calc(ll n, ll x, ll P) {
30     if (!n) return 1;
31     ll s = 1;
32     for (ll i = 1; i <= P; i++)

```

```

33     if (i % x) s = s * i % P;
34     s = qpow(s, n / P, P);
35     for (ll i = n / P * P + 1; i <= n; i++)
36         if (i % x) s = i % P * s % P;
37     return s * calc(n / x, x, P) % P;
38 }
39 ll multilucas(ll n, ll m, ll x, ll P) {
40     if (n < m) return 0;
41     int cnt = 0;
42     for (ll i = n; i; i /= x) cnt += i / x;
43     for (ll i = m; i; i /= x) cnt -= i / x;
44     for (ll i = n - m; i; i /= x) cnt -= i / x;
45     return qpow(x, cnt, P) % P * calc(n, x, P) % P *
46     inv(calc(m, x, P), P) % P *
47     inv(calc(n - m, x, P), P) % P;
48 }
49 ll exlucas(ll n, ll m, ll P) {
50     int cnt = 0;
51     ll p[20], a[20];
52     for (ll i = 2; i * i <= P; i++) {
53         if (P % i == 0) {
54             p[++cnt] = i;
55             while (P % i == 0) p[cnt] = p[cnt] * i, P /= i;
56             a[cnt] = multilucas(n, m, i, p[cnt]);
57         }
58     }
59     if (P > 1) p[++cnt] = P, a[cnt] = multilucas(n, m, P, P);
60     return CRT(cnt, a, p);
}

```

13.12 Divisor Block

b5dde6

```

1 // calls f(l, r, q) for each maximal [l, r] with floor(n/i)
  = q
2 void divisorBlock(ll n, auto f) {
3     for (ll l = 1, r, q; l <= n; l = r + 1) {
4         q = n / l, r = n / q;
5         f(l, r, q);
6     }
7 /* Example: sum of floor(n/i), i from 1 to n
8 ll ans = 0;
9 divisorBlock(n, [&](ll l, ll r, ll q) {
10     ans += q * (r - l + 1);
11 }); */

```

13.13 Formulas

- Pisano Period: 任何線性遞迴 (比如費氏數列) 模任何一個數字 M 都會循環, 找循環節 $\pi(M)$ 先質因數分解 $M = \prod p_i^{e_i}$, 然後 $\pi(M) = \text{lcm}(\pi(p_i^{e_i}))$
- Inversion: $aa^{-1} \equiv 1 \pmod{m}$, exists iff $\gcd(a, m) = 1$. Linear: $a^{-1} \equiv (m - \lfloor \frac{m}{a} \rfloor) \times (m \bmod a)^{-1} \pmod{m}$
- Fermat: $a^p \equiv a \pmod{p}$; Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$. Not coprime: 分解 $n = \prod p_i^{e_i}$, 對每個 $p_i^{e_i}$ 分開看 (互質: Euler / 不互質: 指數 $\geq e_i$ 直接是 0), 最後 CRT 合併. Also $a^b \equiv a^{b \bmod \varphi(n) + \varphi(n)} \pmod{n}$ for $b \geq \log_2 n$.
- Wilson: $(p-1)! \equiv -1 \pmod{p}$
- $\varphi(n) = n \prod_{p|n} (1 - \frac{1}{p})$, $\sum_{d|n} \varphi(d) = n$
- Divisor function: $n = \prod p_i^{a_i}$, $\sigma_x(n) = \prod \frac{p_i^{(a_i+1)x} - 1}{p_i^x - 1}$ ($x \neq 0$), $\sigma_0(n) = \prod (a_i + 1)$
- CRT (coprime): $M = \prod m_i$, $M_i = \frac{M}{m_i}$, $x = \sum a_i M_i (M_i^{-1} \bmod m_i) \pmod{M}$
General: $x = m_1 p + a_1 = m_2 q + a_2 \Rightarrow m_1 p - m_2 q = a_2 - a_1$, solve by extgcd, answer mod $\text{lcm}(m_1, m_2)$
- Avoiding overflow: $ca \bmod cb = c(a \bmod b)$
- Dirichlet convolution: $(f * g)(n) = \sum_{d|n} f(d)g(\frac{n}{d})$. $\varepsilon(n) = [n=1]$, $1(n) = 1$, $id(n) = n$; $\mu * 1 = \varepsilon$, $\varphi = \mu * id$, $id = \varphi * 1$, $\sigma_0 = 1 * 1$, $\sigma_1 = id * 1$
- $[\gcd(a, b) = 1] = \sum_{d|\gcd(a, b)} \mu(d)$; Möbius inversion: $f = g * 1 \Leftrightarrow g = f * \mu$
- $\sum_{i=1}^n \sum_{j=1}^m [\gcd(i, j) = 1] = \sum_d \mu(d) \lfloor \frac{n}{d} \rfloor \lfloor \frac{m}{d} \rfloor$ (divisor block, $O(\sqrt{n})$ per query)
- Legendre: $v_p(n!) = \sum_{k \geq 1} \lfloor \frac{n}{p^k} \rfloor = \frac{n - s_p(n)}{p-1}$, s_p = digit sum in base p
- Lucas (p prime): $\binom{n}{m} \equiv \prod \binom{n_i}{m_i} \pmod{p}$ over base- p digits; Kummer: $v_p(\binom{n}{m})$ = number of carries adding $m + (n - m)$ in base p

14 Polynomial

14.1 Generating Functions

OGF $A(x) = \sum a_i x^i$: $A(rx) \Rightarrow r^n a_n$; $AB \Rightarrow \sum_i a_i b_{n-i}$; $A^k \Rightarrow \sum_{i_1+\dots+i_k=n} a_{i_1} \dots a_{i_k}$; $x A'(x) \Rightarrow n a_n$; $\frac{A(x)}{1-x} \Rightarrow \sum_{i \leq n} a_i$.
EGF $A(x) = \sum a_i \frac{x^i}{i!}$: $A^{(k)}(x) \Rightarrow a_{n+k}$; $AB \Rightarrow \sum_i \binom{n}{i} a_i b_{n-i}$; $A^k \Rightarrow \sum \frac{n!}{i_1! \dots i_k!} a_{i_1} \dots a_{i_k}$; $x A' \Rightarrow n a_n$.

$$(1+x)^n = \sum \binom{n}{i} x^i, \quad \frac{1}{(1-x)^n} = \sum_i \binom{i+n-1}{n-1} x^i, \quad \frac{1}{1-x} = \sum x^i,$$

$$e^x = \sum \frac{x^i}{i!}, \quad -\ln(1-x) = \sum_{i \geq 1} \frac{x^i}{i}$$

$$\text{Catalan } C(x) = \frac{1-\sqrt{1-4x}}{2x}; \text{Fibonacci } \frac{x}{1-x-x^2}; \sum_i i x^i = \frac{x}{(1-x)^2}; \text{partitions } \prod_k \frac{1}{1-x^k}.$$

14.2 NTT

b4c099

```

1 // mod 998244353, Requirement: ModBasic.cpp
2 const int N = 1 << 20, G = 3;
3 using Poly = vector<ll>;
4
5 struct NTT {
6     ll w[N];
7     NTT() {
8         ll dw = Pow(G, (mod - 1) / N);
9         w[0] = 1;
10        for (int i = 1; i < N; i++)
11            w[i] = mul(w[i-1], dw);
12    }
13    void operator()(Poly& a, bool inv = false) {
14        int n = sz(a);
15        for (int j = 1, x = 0; j < n - 1; j++) {
16            for (int k = n >> 1; (x ^= k) < k; k >>= 1);
17            if (j < x) swap(a[j], a[x]);
18        }
19        for (int L = 2; L <= n; L <= 1) {
20            int dx = N / L, dl = L >> 1;
21            for (int i = 0; i < n; i += L)
22                for (int j = i, x = 0; j < i + dl; j++, x += dx) {
23                    ll tmp = mul(a[j + dl], w[x]);
24                    a[j + dl] = sub(a[j], tmp);
25                    a[j] = add(a[j], tmp);
26                }
27        }
28        if (inv) {
29            reverse(1 + all(a));
30            ll invn = Pow(n, mod - 2);
31            for (ll& x : a) x = mul(x, invn);
32        }
33    } ntt;
34    // polyMul: a*b truncated to bound terms, O(n log n).
35    // Result length must be <= N.
36    Poly polyMul(Poly a, Poly b, int bound = N) {
37        int m = sz(a) + sz(b) - 1, n = 1;
38        while (n < m) n <= 1;
39        a.resize(n), b.resize(n);
40        ntt(a), ntt(b);
41        for (int i = 0; i < n; i++) a[i] = mul(a[i], b[i]);
42        ntt(a, true);
43        a.resize(min(m, bound));
44        return a;
45    }
46    // NTT primes & primitive roots (p = c * 2^k + 1)
47    // max convolution length (|a|+|b|-1) = 2^k
48    // Prime = c * 2^k + 1
49    // 7681 = 15 * 2^9 + 1
50    // 12289 = 3 * 2^12 + 1
51    // 40961 = 5 * 2^13 + 1
52    // 65537 = 1 * 2^16 + 1
53    // 786433 = 3 * 2^18 + 1
54    // 5767169 = 11 * 2^19 + 1
55    // 7340033 = 7 * 2^20 + 1
56    // 23068673 = 11 * 2^21 + 1
57    // 104857601 = 25 * 2^22 + 1
58    // 985661441 = 235 * 2^22 + 1
59    // 998244353 = 119 * 2^23 + 1
60    // 2810183681 = 335 * 2^23 + 1
61    // 167772161 = 5 * 2^25 + 1
62    // 1107296257 = 33 * 2^25 + 1
63    // 469762049 = 7 * 2^26 + 1
64    // 2885681153 = 43 * 2^26 + 1
65    // 2013265921 = 15 * 2^27 + 1
66    // 605028353 = 577 * 2^20 + 1
67    // 2061584302081 = 15 * 2^37 + 1
68    // 2748779069441 = 5 * 2^39 + 1
69    // 194555039024054273 = 27 * 2^56 + 1
70    // 72057594037927936

```

14.3 FFT

a6c438

```

1 // Rounding is safe if
2 // (sum a[i]^2 + sum b[i]^2) log2(N) < 9e14.
3 // In practice, 1e16 or higher often works.
4 using C = complex<double>;
5 using vd = vector<double>;
6
7 void fft(vector<C>& a) {
8     int n = sz(a), L = 31 - __builtin_clz(n);
9     static vector<complex<long double>> R(2, 1);
10    static vector<C> rt(2, 1);

```

```

11 for (static int k = 2; k < n; k *= 2) {
12     R.resize(n), rt.resize(n);
13     auto x = polar(1.0L, acos(-1.0L) / k);
14     for (int i = k; i < 2 * k; i++)
15         rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
16 }
17 vi rev(n);
18 for (int i = 0; i < n; i++)
19     rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
20 for (int i = 0; i < n; i++)
21     if (i < rev[i]) swap(a[i], a[rev[i]]);
22 for (int k = 1; k < n; k *= 2)
23     for (int i = 0; i < n; i += 2 * k)
24         for (int j = 0; j < k; j++) {
25             auto* x = (double*)&rt[j + k];
26             auto* y = (double*)&a[i + j + k];
27             C z(x[0] * y[0] - x[1] * y[1],
28               x[0] * y[1] + x[1] * y[0]);
29             a[i + j + k] = a[i + j] - z;
30             a[i + j] += z;
31         }
32 }
33 vd convFFT(const vd& a, const vd& b) {
34     if (a.empty() || b.empty()) return {};
35     vd res(sz(a) + sz(b) - 1);
36     int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
37     vector<C> in(n), out(n);
38     copy(all(a), begin(in));
39     for (int i = 0; i < sz(b); i++) in[i].imag(b[i]);
40     fft(in);
41     for (C& x : in) x *= x;
42     for (int i = 0; i < n; i++)
43         out[i] = in[-i & (n - 1)] - conj(in[i]);
44     fft(out);
45     for (int i = 0; i < sz(res); i++)
46         res[i] = imag(out[i]) / (4 * n);
47     return res;
48 }
49 }

```

14.4 FFT Mod (arbitrary modulus)

185078

```

1 // Higher precision FFT convolution modulo M.
2 // Safe when N log2(N) M < 8.6e14.
3 // Inputs are in [0,M), with M < 2^31-ish.
4 // needs: FFT
5 vector<ll> convMod(const vector<ll>& a,
6                   const vector<ll>& b, ll M) {
7     if (a.empty() || b.empty()) return {};
8     vector<ll> res(sz(a) + sz(b) - 1);
9     int B = 32 - __builtin_clz(sz(res)), n = 1 << B;
10    int cut = int(sqrt(M));
11    vector<C> L(n), R(n), outs(n), outl(n);
12    for (int i = 0; i < sz(a); i++)
13        L[i] = C((int)(a[i] / cut), (int)(a[i] % cut));
14    for (int i = 0; i < sz(b); i++)
15        R[i] = C((int)(b[i] / cut), (int)(b[i] % cut));
16    fft(L), fft(R);
17    for (int i = 0; i < n; i++) {
18        int j = -i & (n - 1);
19        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
20        outs[j] = (L[i] - conj(L[j])) * R[i]
21                  / (2.0 * n * C(0, 1));
22    }
23    fft(outl), fft(outs);
24    for (int i = 0; i < sz(res); i++) {
25        ll av = ll(real(outl[i]) + .5);
26        ll cv = ll(imag(outs[i]) + .5);
27        ll bv = ll(imag(outl[i]) + .5) + ll(real(outs[i]) + .5);
28        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
29    }
30    return res;
31 }

```

14.5 FWHT

8e386a

```

1 using Poly = vector<ll>;
2
3 // op: 0 = AND, 1 = OR, 2 = XOR; inv = inverse transform
4 // conv: fwt(a,op), fwt(b,op), a[i] = mul(a[i], b[i]),
5 // fwt(a,op,1) => a is the answer
6 void fwt(vector<ll>& a, int op, bool inv = false) {
7     int n = a.size(); ll i2 = (mod + 1) / 2;
8     for (int s = 1; s < n; s <= 1)
9         for (int i = 0; i < n; i += s << 1)
10             for (int j = i; j < i + s; j++) {
11                 ll &x = a[j], &y = a[j + s];
12                 if (op == 0) x = inv ? sub(x, y) : add(x, y);
13                 else if (op == 1) y = inv ? sub(y, x) : add(y, x);
14                 else {
15                     tie(x, y) = pair(add(x, y), sub(x, y));
16                     if (inv) x = mul(x, i2), y = mul(y, i2);

```

```

17         }
18     }
19 }

```

14.6 Polynomial Operations

b9f6d5

```

1 // needs: NTT
2 // needs: ModSqrt
3 // polyInv(a): inverse modulo x^n, a[0] != 0.
4 Poly polyInv(Poly a) {
5     int n = sz(a);
6     Poly res(1, Pow(a[0], mod - 2));
7     for (int m = 1; m < n; m <= 1) {
8         if (n < m * 2) a.resize(m * 2);
9         Poly v1(a.begin(), a.begin() + m * 2), v2 = res;
10        v1.resize(m * 4), v2.resize(m * 4);
11        ntt(v1), ntt(v2);
12        for (int i = 0; i < m * 4; i++)
13            v1[i] = mul(mul(v1[i], v2[i]), v2[i]);
14        ntt(v1, true);
15        res.resize(m * 2);
16        for (int i = 0; i < m; i++) res[i] = add(res[i],
17        res[i]);
18        for (int i = 0; i < m * 2; i++)
19            res[i] = sub(res[i], v1[i]);
20    }
21    res.resize(n);
22    return res;
23 }
24 // polyDiv(a,b): a = bQ + R, b.back() != 0.
25 pair<Poly, Poly> polyDiv(Poly a, Poly b) {
26     int n = sz(a), m = sz(b), k = n - m + 1;
27     if (n < m) return {{0}, a};
28     Poly ra = a, rb = b;
29     reverse(all(ra)), ra.resize(k);
30     reverse(all(rb)), rb.resize(k);
31     Poly q = polyMul(ra, polyInv(rb), k);
32     reverse(all(q));
33     Poly res = polyMul(b, q), r(m - 1);
34     for (int i = 0; i < m - 1; i++) r[i] = sub(a[i], res[i]);
35     return {q, r};
36 }
37
38 // polySqrt(a): square root modulo x^n; asserts if a[0]
39 // not a residue.
40 Poly polySqrt(Poly a) {
41     if (a.empty()) return {0};
42     int n = sz(a), m = 0;
43     while (m < n && a[m] == 0) m++;
44     if (m == n) return Poly(n);
45     if (m & 1) return {-1};
46     a.erase(a.begin(), a.begin() + m);
47     ll z = sqrtMod(a[0], mod);
48     Poly q(1, z);
49     ll inv2 = (mod + 1) / 2;
50     for (int k = 1; k < n - m + 1; k <= 1) {
51         if (n - m < k * 2) a.resize(k * 2);
52         q.resize(k * 2);
53         Poly f2 = polyMul(q, q, k * 2);
54         for (int i = 0; i < k * 2; i++) f2[i] = sub(f2[i],
55         a[i]);
56         f2 = polyMul(f2, polyInv(q), k * 2);
57         for (int i = 0; i < k * 2; i++)
58             q[i] = sub(q[i], mul(f2[i], inv2));
59     }
60     q.resize(n - m);
61     Poly res(n);
62     for (int i = 0; i < sz(q); i++) res[i + m / 2] = q[i];
63     return res;
64 }
65 Poly deriv(Poly a) {
66     int n = sz(a);
67     Poly res(max(0, n - 1));
68     for (int i = 0; i < n - 1; i++)
69         res[i] = mul(a[i + 1], i + 1);
70     return res;
71 }
72
73 Poly integ(Poly a) {
74     int n = sz(a);
75     Poly res(n + 1);
76     for (int i = 0; i < n; i++)
77         res[i + 1] = mul(a[i], Pow(i + 1, mod - 2));
78     return res;
79 }
80
81 // polyLn(a): formal logarithm, a[0] = 1.
82 Poly polyLn(Poly a) {

```



```

83 int n = sz(a);
84 if (n == 1) return {0};
85 Poly d = deriv(a);
86 a.pop_back();
87 return integ(polyMul(d, polyInv(a), n - 1));
88 }
89
90 // polyExp(a): formal exponential, a[0] = 0.
91 Poly polyExp(Poly a) {
92     int n = sz(a);
93     Poly q(1, 1);
94     a[0] = add(a[0], 1);
95     for (int m = 1; m < n; m <= 1) {
96         if (n < m * 2) a.resize(m * 2);
97         Poly g(a.begin(), a.begin() + m * 2), h = q;
98         h.resize(m * 2), h = polyLn(h);
99         for (int i = 0; i < m * 2; i++) g[i] = sub(g[i], h[i]);
100         q = polyMul(g, q, m * 2);
101     }
102     q.resize(n);
103     return q;
104 }
105
106 Poly polyPow(Poly a, ll k) {
107     int n = sz(a), m = 0;
108     Poly ans(n);
109     if (!k) return ans[0] = 1, ans;
110     while (m < n && a[m] == 0) m++;
111     if (k && m && (k >= n || k * m >= n)) return ans;
112     if (m == n) return ans[0] = 1, ans;
113     int lead = m * k;
114     Poly b(a.begin() + m, a.end());
115     ll base = Pow(b[0], k), inv = Pow(b[0], mod - 2);
116     for (int i = 0; i < n - m; i++) b[i] = mul(b[i], inv);
117     b = polyLn(b);
118     for (ll& x : b) x = mul(x, k % mod);
119     b = polyExp(b);
120     for (int i = lead; i < n; i++)
121         ans[i] = mul(b[i - lead], base);
122     return ans;
123 }
124
125 vector<ll> evaluate(Poly a, vector<ll> x) {
126     if (x.empty()) return {};
127     int n = sz(x);
128     vector<Poly> up(n * 2);
129     for (int i = 0; i < n; i++) up[i + n] = {sub(0, x[i]), 1};
130     for (int i = n - 1; i > 0; i--)
131         up[i] = polyMul(up[i * 2], up[i * 2 + 1]);
132     vector<Poly> down(n * 2);
133     down[1] = polyDiv(a, up[1]).second;
134     for (int i = 2; i < n * 2; i++)
135         down[i] = polyDiv(down[i >> 1], up[i]).second;
136     vector<ll> y(n);
137     for (int i = 0; i < n; i++) y[i] = down[i + n][0];
138     return y;
139 }
140
141 Poly interpolate(vector<ll> x, vector<ll> y) {
142     int n = sz(x);
143     vector<Poly> up(n * 2);
144     for (int i = 0; i < n; i++) up[i + n] = {sub(0, x[i]), 1};
145     for (int i = n - 1; i > 0; i--)
146         up[i] = polyMul(up[i * 2], up[i * 2 + 1]);
147     Poly a = evaluate(deriv(up[1]), x);
148     for (int i = 0; i < n; i++)
149         a[i] = mul(y[i], Pow(a[i], mod - 2));
150     vector<Poly> down(n * 2);
151     for (int i = 0; i < n; i++) down[i + n] = {a[i], 1};
152     for (int i = n - 1; i > 0; i--) {
153         Poly lhs = polyMul(down[i * 2], up[i * 2 + 1]);
154         Poly rhs = polyMul(down[i * 2 + 1], up[i * 2]);
155         down[i].resize(sz(lhs));
156         for (int j = 0; j < sz(lhs); j++)
157             down[i][j] = add(lhs[j], rhs[j]);
158     }
159     return down[1];
160 }

```

14.7 Berlekamp Massey

b65456

```

1 // Returns c with s[i] = \sum c[j] * s[i-1-j].
2 // O(n^2); you should provide at least 2k samples for order k.
3 // Require: ModBasic.cpp
4 using Poly = vector<ll>;
5 Poly berlekampMassey(Poly s) {
6     int n = s.size(), L = 0, m = 0;
7     Poly C(n), B(n), T;
8     C[0] = B[0] = 1; ll b = 1;
9     for (int i = 0; i < n; i++) {

```

```

10     m++; ll d = s[i] % mod;
11     for (int j = 1; j <= L; j++)
12         d = add(d, mul(C[j], s[i - j]));
13     if (!d) continue;
14     T = C;
15     ll coef = mul(d, Pow(b, mod - 2));
16     for (int j = m; j < n; j++)
17         C[j] = sub(C[j], mul(coef, B[j - m]));
18     if (2 * L > i) continue;
19     L = i + 1 - L;
20     B = T, b = d, m = 0;
21 }
22 C.resize(L + 1);
23 C.erase(C.begin());
24 for (ll& x : C) x = x % mod - x < 0 ? mod - x : 0;
25 return C;
26 }
27 /* Example: Fibonacci
28 Poly s = {0, 1, 1, 2, 3, 5, 8, 13};
29 Poly c = berlekampMassey(s); // c = {1, 1}, k = 2
30 ll ans = linearRec(s, c, (ll)1e18); // find fib(1e18) */

```

14.8 Linear Recurrence (Kitamasa)

496400

```

1 // k-th term of a recurrence in O(n^2 log k)
2 // S[i] = \sum C[j] * S[i-1-j]
3 // Example: linearRec({0,1},{1,1},k) returns Fib(k).
4 // Require: ModBasic.cpp
5 using Poly = vector<ll>;
6 ll linearRec(Poly S, Poly C, ll k) {
7     int n = C.size();
8     auto combine = [&](Poly a, Poly b) {
9         Poly res(n * 2 + 1);
10         for (int i = 0; i <= n; i++)
11             for (int j = 0; j <= n; j++)
12                 res[i + j] = add(res[i + j], mul(a[i], b[j]));
13         for (int i = 2 * n; i > n; i--)
14             for (int j = 0; j < n; j++)
15                 res[i - 1 - j] = add(res[i - 1 - j],
16                                     mul(res[i], C[j]));
17         res.resize(n + 1);
18         return res;
19     };
20     Poly pol(n + 1, e(pol));
21     pol[0] = e[1] = 1;
22     for (++k; k; k /= 2) {
23         if (k % 2) pol = combine(pol, e);
24         e = combine(e, e);
25     }
26     ll res = 0;
27     for (int i = 0; i < n; i++)
28         res = add(res, mul(pol[i + 1], S[i]));
29     return res;
30 }

```

14.9 Lagrange Interpolation (iota points)

8075cc

```

1 Mint lagrange_iota(const vector<Mint> &f, const int c) {
2     const int n = int(size(f));
3     if (c < n) {
4         return f[c];
5     }
6     vector<Mint> fac(n), ifac(n);
7     fac[0] = 1;
8     for (int i = 1; i < n; i++) fac[i] = fac[i - 1] * i;
9     ifac[n - 1] = fac[n - 1].inv();
10    for (int i = n - 1; i >= 1; i--) ifac[i - 1] = ifac[i] * i;
11    vector<Mint> g(n);
12    for (int i = 0; i < n; i++) {
13        g[i] = f[i] * ifac[i] * ifac[n - 1 - i];
14        if ((n - 1 - i) & 1) g[i] *= -1;
15    }
16    vector<Mint> lp(n + 1), rp(n + 1);
17    lp[0] = rp[n] = 1;
18    for (int i = 0; i < n; i++) lp[i + 1] = lp[i] * Mint(c - i);
19    for (int i = n - 1; i >= 0; i--) rp[i] = rp[i + 1] * Mint(c - i);
20    Mint ans = 0;
21    for (int i = 0; i < n; i++) ans += g[i] * lp[i] * rp[i + 1];
22    return ans;
23 }

```

15 Linear Algebra

15.1 Gauss (mod p)

dad24a

```

1 // In-place RREF over Z_mod (mod prime); returns {rank, det}.
2 // det is valid only when A is square.
3 // Values must be normalized.
4 pair<int, ll> gauss(vector<vector<ll>>&A) {

```

```

5 int n = A.size(), m = A[0].size();
6 int rank = 0; ll det = 1;
7 for (int c = 0; c < m && rank < n; c++) {
8     int piv = rank;
9     while (piv < n && !A[piv][c]) piv++;
10    if (piv == n) { det = 0; continue; }
11    if (piv != rank) {
12        swap(A[piv], A[rank]);
13        det = sub(0, det);
14    }
15    ll inv = Pow(A[rank][c], mod - 2);
16    det = mul(det, A[rank][c]);
17    for (ll& x : A[rank]) x = mul(x, inv);
18    for (int i = 0; i < n; i++) if (i != rank && A[i][c]) {
19        ll f = A[i][c];
20        for (int j = c; j < m; j++)
21            A[i][j] = sub(A[i][j], mul(f, A[rank][j]));
22    }
23    rank++;
24 }
25 return {rank, det};
26 }

```

15.2 Solve Linear Equations

07cfff4

```

1 // Author: Gino
2 // solve(Augmented Matrix), nullity == -1 if no solution
3 struct Solution {
4     int nullity; vector<ll> xp; vector<vector<ll>> basis;
5 };
6 Solution solve(vector<vector<ll>>& A) { // augmented matrix
7     int n = A.size(), m = A[0].size() - 1;
8     auto [rank, det] = gauss(A);
9     for (int i = 0; i < n; i++) if (A[i][m] != 0) {
10         if (ranges::all_of(A[i] | views::take(m),
11             [](ll x){ return x == 0; }))
12             return {-1, {}, {}};
13     }
14     Solution sol;
15     vector<int> pcol(rank); // pivot columns
16     vector<bool> isp(m, false); // is the column a pivot?
17     for (int i = 0; i < rank; i++) {
18         auto it = ranges::find(A[i], 1);
19         pcol[i] = ranges::distance(A[i].begin(), it);
20         isp[pcol[i]] = true;
21     }
22     sol.nullity = m - rank;
23     sol.xp.assign(m, 0);
24     for (int i = 0; i < rank; i++) sol.xp[pcol[i]] = A[i][m];
25     for (int j = 0; j < m; j++) if (!isp[j]) {
26         vector<ll> d(m, 0); d[j] = 1;
27         for (int i = 0; i < rank; i++) d[pcol[i]] = sub(0, A[i][j]);
28         sol.basis.push_back(d);
29     }
30     return sol;
31 }

```

15.3 Gauss (GF(2) / XOR basis)

c8d703

```

1 // In-place RREF over GF(2); rows are bitsets, m = #columns.
2 // Returns {rank, det}; det is valid only when A is square.
3 // For an XOR basis, run it and read the nonzero rows.
4 const int N = 1000;
5 using bs = bitset<N>;
6 pair<int, int> gaussBinary(vector<bs>& A, int m) {
7     int n = A.size(), rank = 0;
8     for (int c = 0; c < m && rank < n; c++) {
9         int piv = rank;
10        while (piv < n && !A[piv][c]) piv++;
11        if (piv == n) continue;
12        swap(A[piv], A[rank]);
13        for (int i = 0; i < n; i++)
14            if (i != rank && A[i][c]) A[i] ^= A[rank];
15        rank++;
16    }
17    return {rank, n == m && rank == n};
18 }

```

15.4 Solve Linear Equations (GF(2))

7324a9

```

1 // solveBinary(A, m): A is augmented (column m is b),
2 // m = #variables. nullity == -1 if no solution.
3 struct BinSolution { int nullity; bs xp; vector<bs> basis; };
4 BinSolution solveBinary(vector<bs>& A, int m) {
5     int n = A.size();
6     auto [rank, det] = gaussBinary(A, m + 1);
7     for (int i = 0; i < n; i++) // row: 0 ... 0 | 1
8         if ((int)A[i].Find_first() == m) return {-1, {}, {}};
9     BinSolution sol{m - rank, {}, {}};
10    vector<int> pcol(rank); // pivot columns
11    bs isp; // is the column a pivot?

```

```

12 for (int i = 0; i < rank; i++) {
13     pcol[i] = A[i].Find_first();
14     isp[pcol[i]] = 1; sol.xp[pcol[i]] = A[i][m];
15 }
16 for (int j = 0; j < m; j++) if (!isp[j]) {
17     bs d; d[j] = 1;
18     for (int i = 0; i < rank; i++) d[pcol[i]] = A[i][j];
19     sol.basis.push_back(d);
20 }
21 return sol;
22 }

```

15.5 Determinant

- Row swap: $-\det$; $k\vec{r}_i$: $k\det$; $\vec{r}_i + k\vec{r}_j$: unchanged; $\det(A^T) = \det(A)$, $\det(AB) = \det(A)\det(B)$.
- Matrix-tree: number of spanning trees = any cofactor of Laplacian $D - A$. Directed (arborescences rooted at r): delete row/col r of $D_{in} - A$.
- Cayley: n^{n-2} labeled trees; forests of k rooted trees on given roots: kn^{n-k-1} .

16 Numerical

16.1 Integrate (Simpson / adaptive)

6fe0f0

```

1 // Simpson integration has fourth-order error
2 // in the step size.
3 template<class F>
4 double simpsonPart(double a, double b, F& f) {
5     double c = (a + b) / 2;
6     return (f(a) + 4 * f(c) + f(b)) * (b - a) / 6;
7 }
8
9 template<class F>
10 double simpson(double a, double b, F f, int n = 1000) {
11     double h = (b - a) / (2 * n);
12     double v = f(a) + f(b);
13     for (int i = 1; i < 2 * n; i++)
14         v += f(a + i*h) * (i&1 ? 4 : 2);
15     return v * h / 3;
16 }
17
18 // Adaptive Simpson integration to absolute error eps.
19 template<class F>
20 double adaptiveSimpson(F& f, double a, double b,
21     double eps, double whole) {
22     double c = (a + b) / 2;
23     double left = simpsonPart(a, c, f);
24     double right = simpsonPart(c, b, f);
25     double sum = left + right;
26     if (abs(sum - whole) <= 15 * eps ||
27         abs(b - a) < 1e-10)
28         return sum + (sum - whole) / 15;
29     return adaptiveSimpson(f, a, c, eps / 2, left) +
30         adaptiveSimpson(f, c, b, eps / 2, right);
31 }
32
33 template<class F>
34 double quad(double a, double b, F f, double eps = 1e-8) {
35     return adaptiveSimpson(f, a, b, eps, simpsonPart(a, b, f));
36 }

```

16.2 Linear Programming

Standard form: $\max c^T x$ s.t. $Ax \leq b, x \geq 0$. Dual: $\min b^T y$ s.t. $A^T y \geq c, y \geq 0$; optima equal (strong duality). Complementary slackness: $\bar{x}_i \bar{y}_j$ optimal iff for all i : $\bar{x}_i = 0$ or $\sum_j A_{ji} \bar{y}_j = c_i$, and for all j : $\bar{y}_j = 0$ or $\sum_i A_{ji} \bar{x}_i = b_j$. To standard form: minimize $\Rightarrow$ negate c ; $\geq$ row $\Rightarrow$ negate row; $=$ row $\Rightarrow$ both $\leq$ and $\geq$; free $x_i \Rightarrow x_i - x_i$.

16.3 Simplex

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```

1 // Maximize c^T x subject to Ax <= b, x >= 0.
2 // Returns -inf if infeasible, inf if unbounded.
3 // Otherwise returns the maximum value and stores x.
4 // Numerical stability is not guaranteed.
5 using T = double;
6 using vd = vector<T>;
7 using vvd = vector<vd>;
8
9 const T eps = 1e-8;
10 const T inf = numeric_limits<T>::infinity();
11
12 struct LPSolver {
13     int m, n;
14     vi N, B;
15     vvd D;
16
17     LPSolver(const vvd& A, const vd& b, const vd& c)
18         : m(sz(b)), n(sz(c)), N(n + 1), B(m),
19           D(m + 2, vd(n + 2)) {}
20     for (int i = 0; i < m; i++)

```

```

21     for (int j = 0; j < n; j++) D[i][j] = A[i][j];
22     for (int i = 0; i < m; i++) {
23         B[i] = n + i;
24         D[i][n] = -1;
25         D[i][n + 1] = b[i];
26     }
27     for (int j = 0; j < n; j++) D[m][j] = -c[j];
28     N[n] = -1;
29     D[m + 1][n] = 1;
30 }
31
32 void pivot(int r, int s) {
33     T* a = D[r].data();
34     T inv = 1 / a[s];
35     for (int i = 0; i < m + 2; i++) if (i != r &&
36         abs(D[i][s]) > eps) {
37         T* b = D[i].data();
38         T inv2 = b[s] * inv;
39         for (int j = 0; j < n + 2; j++) b[j] -= a[j] * inv2;
40         b[s] = a[s] * inv2;
41     }
42     for (int j = 0; j < n + 2; j++)
43         if (j != s) D[r][j] *= inv;
44     for (int i = 0; i < m + 2; i++)
45         if (i != r) D[i][s] *= -inv;
46     D[r][s] = inv;
47     swap(B[r], N[s]);
48 }
49
50 bool simplex(int phase) {
51     int x = m + phase - 1;
52     for (;;) {
53         int s = -1;
54         for (int j = 0; j <= n; j++) {
55             if (N[j] == -phase) continue;
56             if (s == -1 || pair(D[x][j], N[j]) <
57                 pair(D[x][s], N[s])) s = j;
58         }
59         if (D[x][s] >= -eps) return true;
60         int r = -1;
61         for (int i = 0; i < m; i++) {
62             if (D[i][s] <= eps) continue;
63             if (r == -1 || pair(D[i][n + 1] / D[i][s], B[i]) <
64                 pair(D[r][n + 1] / D[r][s], B[r])) r = i;
65         }
66         if (r == -1) return false;
67         pivot(r, s);
68     }
69 }
70
71 T solve(vd& x) {
72     int r = 0;
73     for (int i = 1; i < m; i++)
74         if (D[i][n + 1] < D[r][n + 1]) r = i;
75     if (D[r][n + 1] < -eps) {
76         pivot(r, n);
77         if (!simplex(2) || D[m + 1][n + 1] < -eps) return -
78     inf;
79     for (int i = 0; i < m; i++) if (B[i] == -1) {
80         int s = 0;
81         for (int j = 1; j <= n; j++)
82             if (s == 0 || pair(D[i][j], N[j]) <
83                 pair(D[i][s], N[s])) s = j;
84         pivot(i, s);
85     }
86     bool ok = simplex(1);
87     x = vd(n);
88     for (int i = 0; i < m; i++)
89         if (B[i] < n) x[B[i]] = D[i][n + 1];
90     return ok ? D[m][n + 1] : inf;
91 }
92 };

```

17 Combinatorics

17.1 Catalan Number

$$C_0 = 1, C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}, C_n = C_n^{2n} - C_{n-1}^{2n}$$

0	1	1	2	5
4	14	42	132	429
8	1430	4862	16796	58786
12	208012	742900	2674440	9694845

17.2 Bertrand's Ballot Theorem

- A always $> B$: $C(p+q, p) - 2C(p+q-1, p)$

- A always $\geq B$: $C(p+q, p) \times \frac{p+1-q}{p+1}$

17.3 Burnside's Lemma

Let X be the original set.

Let G be the group of operations acting on X .

Let X^g be the set of x not affected by g .

Let X/G be the set of orbits. Then the following equation holds:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

17.4 Binomial Identities

- Vandermonde: $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$; $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$;
- $\sum_{k=0}^n \binom{n}{k} \binom{n-k}{b} = \binom{n+1}{a+b+1}$;
- Hockey stick: $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$; $\sum_k k \binom{n}{k} = n2^{n-1}$; $\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$
- Stars and bars: $x_1 + \dots + x_k = n, x_i \geq 0$: $\binom{n+k-1}{k-1}$
- Catalan: $C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}$; paths from $(0, 0)$ to (a, b) never above $y = x$ ($a \geq b$): $\binom{a+b}{b} - \binom{a+b}{b-1}$

17.5 Stirling Numbers

- 1st kind (permutations of n with k cycles): $c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$; $\sum_k c(n, k)x^k = x(x+1)\dots(x+n-1)$
- 2nd kind (partitions of n into k nonempty sets): $S(n, k) = S(n-1, k-1) + kS(n-1, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$ (a convolution, one row in $O(n \log n)$)
- $x^n = \sum_k S(n, k)x^k$, $x^k = x(x-1)\dots(x-k+1)$; Bell $B_n = \sum_k S(n, k)$
- Surjections $n \rightarrow k$: $k!S(n, k)$

17.6 Derangement / Inclusion-Exclusion

- $D_n = (n-1)(D_{n-1} + D_{n-2}) = nD_{n-1} + (-1)^n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$; $D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44$
- $|U \cup A_i| = \sum_{\emptyset \neq S} (-1)^{|S|+1} |\cap_{i \in S} A_i|$; "exactly k of n properties": $\sum_{j \geq k} (-1)^{j-k} \binom{j}{k} N_j$ where N_j = sum over j -subsets of $| \cap |$
- Binomial inversion: $f(n) = \sum_k \binom{n}{k} g(k) \Leftrightarrow g(n) = \sum_k (-1)^{n-k} \binom{n}{k} f(k)$

17.7 Pólya (necklaces)

- Colorings of n -cycle with m colors under rotation: $\frac{1}{n} \sum_d \mid n \varphi(d) m^{\frac{n}{d}}$; with reflection too (n odd): add $nm^{\frac{n+1}{2}}$, (n even): add $\frac{n}{2}(m^{\frac{n}{2}} + m^{\frac{n}{2}+1})$, then divide by $2n$ instead of n .

18 Game Theory

18.1 Nim

- n piles $a_1, \dots, a_n$; 兩人輪流選一堆拿走 ≥ 1 顆，拿不了的人輸 (normal play)。
- 先手必勝 $\Leftrightarrow X = a_1 \oplus \dots \oplus a_n \neq 0$ 。必勝走法：找一堆 a_i 使 $a_i \oplus X < a_i$ (即 a_i 在 X 最高位為 1)，拿成 $a_i \oplus X$ ，讓 XOR 歸零。
- Misère (拿到最後一顆的人輸)：若所有 $a_i \leq 1$ 則勝負相反 (堆數為偶數先手勝)，否則判定同 normal。
- 變形：每次最多拿 k 顆 $\Rightarrow$ 用 $a_i \bmod (k+1)$ 代入；Staircase Nim：只看奇數階的 XOR。

18.2 Sprague-Grundy

- $SG(s) = \text{mex}\{SG(t_1), SG(t_2), \dots\}$ t_i 為對 s 做一步操作後的狀態；終止態 $SG = 0$ (mex = 最小未出現的非負整數)。
- $SG(s) \neq 0 \Leftrightarrow$ 先手必勝；必勝走法：走到任一 $SG(t) = 0$ 的 t 。
- 多個獨立子遊戲同時進行 (每步只能動一個)： $SG = \oplus SG(s_i)$ ；Nim 一堆 a 就是 $SG = a$ 。
- 一步可把一局分裂成多局 (如 Kayles)： t 的 SG 是各分裂部分的 XOR，再取 mex。
- 狀態小時直接打表 $O(|S| \cdot \text{moves})$ ，大狀態先打表找週期／規律。

19 Special Numbers

19.1 Fibonacci Series

1	1	1	2	3
5	5	8	13	21
9	34	55	89	144
13	233	377	610	987
17	1597	2584	4181	6765
21	10946	17711	28657	46368
25	75025	121393	196418	317811
29	514229	832040	1346269	2178309
33	3524578	5702887	9227465	14930352

$$f(45) \approx 10^9, f(88) \approx 10^{18}$$

19.2 Prime Numbers

- $\pi(n) \equiv \text{Number of primes } \leq n \approx \frac{n}{(\ln n)-1}$
 $\pi(100) = 25, \pi(200) = 46$
 $\pi(500) = 95, \pi(1000) = 168$
 $\pi(2000) = 303, \pi(4000) = 550$
 $\pi(10^4) = 1229, \pi(10^5) = 9592$
 $\pi(10^6) = 78498, \pi(10^7) = 664579$

19.3 Number of Divisors

- If $n = \prod p_i^{a_i}$, then $\tau(n) = \prod (a_i + 1)$
- Maximum $\tau(n)$ for $n \leq 10^k$:

k	$\max \tau(n)$	k	$\max \tau(n)$	k	$\max \tau(n)$
3	32	9	1344	15	26880
4	64	10	2304	16	41472
5	128	11	4032	17	64512
6	240	12	6720	18	103680
7	448	13	10752		
8	768	14	17280		

- Useful bounds: $n \leq 10^6 \Rightarrow \tau(n) \leq 240$
 $n \leq 10^9 \Rightarrow \tau(n) \leq 1344$
 $n \leq 10^{12} \Rightarrow \tau(n) \leq 6720$
 $n \leq 10^{15} \Rightarrow \tau(n) \leq 26880$
 $n \leq 10^{18} \Rightarrow \tau(n) \leq 103680$

19.4 Distinct Prime Factors

- Let $\omega(n)$ = number of distinct prime factors.
- Minimum number with k distinct prime factors is the product of the first k primes.
 - $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 = 9699690$
 - $2 \cdot 3 \cdot \dots \cdot 23 = 223092870$
 - $2 \cdot 3 \cdot \dots \cdot 29 = 6469693230$
 - $n \leq 10^9 \Rightarrow \omega(n) \leq 9, n \leq 10^{18} \Rightarrow \omega(n) \leq 15$
- Number of square-free divisors = $2^{\omega(n)}$
 - $n \leq 10^9 \Rightarrow \leq 512, n \leq 10^{18} \Rightarrow \leq 32768$